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## CANCER TREATMENT EFFICACY EVALUATION UPON UNDERGOING OR COMPLETING A CANCER TREATMENT

### Abstract


A computer system is disclosed that evaluates efficacy of a cancer treatment in a subject undergoing or having completed treatment. The system executes a method comprising obtaining first genomic epigenetic sequencing data of the cell-free DNA in a blood sample of the subject. The method further obtains second genomic sequencing data of the cell-free DNA. The first and second data is mapped to locations in a reference genome. Absence or presence of at least a first genomic alteration, in a plurality of genomic alterations, is determined based on at least the second data and the mapped locations of the second data. A structured data report is securely communicated, through a network connection, providing an assessment of the cancer treatment efficacy for the subject responsive to at least the epigenetic patterns of the cell-free DNA and the absence or presence of at least the first genomic alteration.



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## Background/Summary

CROSS REFERENCE TO RELATED PATENT APPLICATIONS [0001] This patent application is a continuation of U.S. patent application Ser. No. 18/298,371, filed Apr. 11, 2023, published as United States Patent Publication No. 2023/0245788 A1, which is a continuation of U.S. Pat. No. 11,640,859, which is a U.S. national stage patent application, filed pursuant to 35 U.S.C. § 371, of International patent application No. PCT/US2019/056713, filed on Oct. 17, 2019, published as WO2020/081795, which claims the benefit of priority of U.S. Provisional patent application No. 62/746,997, filed Oct. 17, 2018, and titled “Data Based Cancer Research and Treatment Systems and Methods, each of which is hereby incorporated by reference.

BACKGROUND OF THE DISCLOSURE
[0002] The present invention relates to systems and methods for obtaining and employing data related to physical and genomic patient characteristics as well as diagnosis, treatments and treatment efficacy to provide a suite of tools to healthcare providers, researchers and other interested parties enabling those entities to develop new cancer state-treatment-results insights and/or improve overall patient healthcare and treatment plans for specific patients.
[0003] The present disclosure is described in the context of a system related to cancer research, diagnosis, treatment and results analysis. Nevertheless, it should be appreciated that the present disclosure is intended to teach concepts, features and aspects that will be useful in many different health related contexts and therefore the specification should not be considered limited to a cancer related systems unless specifically indicated for some system aspect.
[0004] Hereafter, unless indicated otherwise, the following terms and phrases will be used in this disclosure as described. The term “provider” will be used to refer to an entity that operates the overall system disclosed herein and, in most cases, will include a company or other entity that runs servers and maintains databases and that employs people with many different skill sets required to construct, maintain and adapt the disclosed system to accommodate new data types, new medical and treatment insights, and other needs. Exemplary provider employees may include researchers, data abstractors, physicians, pathologists, radiologists, data scientists, and many other persons with specialized skill sets.
[0005] The term “physician” will be used to refer generally to any health care provider including but not limited to a primary care physician, a medical specialist, a physician, a nurse, a medical assistant, etc.
[0006] The term “researcher” will be used to refer generally to any person that performs research including but not limited to a pathologist, a radiologist, a physician, a data scientist, or some other health care provider. One person may operate both a physician and a researcher while others may simply operate in one of those capacities.
[0007] The phrase “system specialist” will be used generally to refer to any provider employee that operates within the disclosed systems to collect, develop, analyze or otherwise process system data,

tissue samples or other information types (e.g., medical images) to generate any intermediate system work product or final work product where intermediate work product includes any data set, conclusions, tissue or other samples, grown tissues or samples, or other information for consumption by one or more other system specialists and where final work product includes data, conclusions or other information that is placed in a final or conclusory report for a system client or that operates within the system to perform research, to adapt the system to changing needs, data types or client requirements. For instance, the phrase “abstractor specialist” will be used to refer to a person that consumes data available in clinical records provided by a physician to generate normalized and structured data for use by other system specialists, the phrase “programming specialist” will be used to refer to a person that generates or modifies application program code to accommodate new data types and or clinical insights, etc.

[0008] The phrase “system user” will be used generally to refer to any person that uses the disclosed system to access or manipulate system data for any purpose and therefore will generally include physicians and researchers that work for the provider or that partner with the provider to perform services for patients or for other partner research institutions as well as system specialists that work for the provider.

[0009] The phrase “cancer state” will be used to refer to a cancer patient's overall condition including diagnosed cancer, location of cancer, cancer stage, other cancer characteristics (e.g., tumor characteristics), other user conditions (e.g., age, gender, weight, race, habits (e.g., smoking, drinking, diet)), other pertinent medical conditions (e.g., high blood pressure, dry skin, other diseases, etc.), medications, allergies, other pertinent medical history, current side effects of cancer treatments and other medications, etc.

[0010] The term “consume” will be used to refer to any type of consideration, use, modification, or other activity related to any type of system data, tissue samples, etc., whether or not that consumption is exhaustive (e.g., used only once, as in the case of a tissue sample that cannot be reproduced) or inexhaustible so that the data, sample, etc., persists for consumption by multiple entities (e.g., used multiple times as in the case of a simple data value).

[0011] The term “consumer” will be used to refer to any system entity that consumes any system data, samples, or other information in any way including each of specialists, physicians, researchers, clients that consume any system work product, and software application programs or operational code that automatically consume data, samples, information or other system work product independent of any initiating human activity.

[0012] The phrase “treatment planning process” will be used to refer to an overall process that includes one or more sub-processes that process clinical and other patient data and samples (e.g., tumor tissue) to generate intermediate data deliverables and eventually final work product in the form of one or more final reports provided to system clients. These processes typically include varying levels of exploration of treatment options for a patient's specific cancer state but are typically related to treatment of a specific patient as opposed to more general exploration for the purpose of more general research activities. Thus, treatment planning may include data generation and processes used to generate that data, consideration of different treatment options and effects of those options on patient illness, etc., resulting in ultimate prescriptive plans for addressing specific patient ailments.

[0013] Medical treatment prescriptions or plans are typically based on an understanding of how treatments affect illness (e.g., treatment results) including how well specific treatments eradicate illness, duration of specific treatments, duration of healing processes associated with specific treatments and typical treatment specific side effects. Ideally treatments result in complete elimination of an illness in a short period with minimal or no adverse side effects. In some cases cost is also a consideration when selecting specific medical treatments for specific ailments.

[0014] Knowledge about treatment results is often based on analysis of empirical data developed over decades or even longer time periods during which physicians and/or researchers have recorded treatment results for many different patients and reviewed those results to identify generally successful ailment specific treatments. Researchers and physicians give medicine to patients or treat an ailment in some other fashion, observe results and, if the results are good, the researchers and physicians use the treatments again to treat similar ailments. If treatment results are bad, a researcher foregoes prescribing the associated treatment for a next encountered similar ailment and instead tries some other treatment, hopefully based on prior treatment efficacy data. Treatment results are sometimes published in medical journals and/or periodicals so that many physicians can benefit from a treating physician's insights and treatment results.

[0015] In many cases treatment results for specific illnesses vary for different patients. In particular, in the case of cancer treatments and results, different patients often respond differently to identical or similar treatments. Recognizing that different patients experience different results given effectively the same treatments in some cases, researchers and physicians often develop additional guidelines around how to optimize ailment treatments based on specific patient cancer state. For instance, while a first treatment may be best for a young relatively healthy woman suffering colon cancer, a second treatment associated with fewer adverse side effects may be optimal for an older relatively frail man with a similar colon same cancer diagnosis. In many cases patient conditions related to cancer state may be gleaned from clinical medical records, via a medical examination and/or via a patient interview, and may be used to develop a personalized treatment plan for a patient's specific cancer state. The idea here is to collect data on as many factors as possible that have any cause-effect relationship with treatment results and use those factors to design optimal personalized treatment plans.

[0016] In treatment of at least some cancer states, treatment and results data is simply inconclusive. To this end, in treatment of some cancer states, seemingly indistinguishable patients with similar conditions often react differently to similar treatment plans so that there is no cause and effect between patient conditions and disparate treatment results. For instance, two women may be the same age, indistinguishably physically fit and diagnosed with the same exact cancer state (e.g., cancer type, stage, tumor characteristics, etc.). Here, the first woman may respond to a cancer treatment plan well and may recover from her disease completely in 8 months with minimal side effects while the second woman, administered the same treatment plan, may suffer several severe adverse side effects and may never fully recover from her diagnosed cancer. Disparate treatment results for seemingly similar cancer states exacerbate efforts to develop treatment and results data sets and prescriptive activities. In these cases, unfortunately, there are cancer state factors that have cause and effect relationships to specific treatment results that are simply currently unknown and therefore those factors cannot be used to optimize specific patient treatments at this time.

[0017] Genomic sequencing has been explored to some extent as another cancer state factor (e.g., another patient condition) that can affect cancer treatment efficacy. To this end, at least some studies have shown that genetic features (e.g., DNA related patient factors (e.g., DNA and DNA alterations) and/or DNA related cancerous material factors (e.g., DNA of a tumor)) as well as RNA and other genetic sequencing data can have cause and effect relationships with at least some cancer treatment results for at least some patients. For instance, in one chemotherapy study using SULT1A1, a gene known to have many polymorphisms that contribute to a reduction of enzyme activity in the metabolic pathways that process drugs to fight breast cancer, patients with a SULT1A1 mutation did not respond optimally to tamoxifen, a widely used treatment for breast cancer. In some cases these patients were simply resistant to the drug and in others a wrong dosage was likely lethal. Side effects ranged in severity depending on varying abilities to metabolize tamoxifen. Raftogianis et al., 1999, *The role of pharmacogenetics in cancer therapy, prevention and risk*, Medical Science Division 243-247. Other cases where genetic features of a patient and/or a tumor affect treatment efficacy are well known.

[0018] While corollaries between genomic features and treatment efficacy have been shown in a small number of cases, it is believed that there are likely many more genomic features and treatment results cause and effect relationships that have yet to be discovered. Despite this belief, genetic testing in cancer cases is the rare exception, not the norm, for several reasons. One problem with genetic testing is that testing is expensive and has been cost prohibitive in many cases.

[0019] Another problem with genetic testing for treatment planning is that, as indicated above, cause and effect relationships have only been shown in a small number of cases and therefore, in most cancer cases, if genetic testing is performed, there is no linkage between resulting genetic factors and treatment efficacy. In other words, in most cases how genetic test results can be used to prescribe better treatment plans for patients is unknown so the extra expense associated with genetic testing in specific cases cannot be justified. Thus, while promising, genetic testing as part of first-line cancer treatment planning has been minimal or sporadic at best.

[0020] While the lack of genetic and treatment efficacy data makes it difficult to justify genetic testing for most cancer patients, perhaps the greater problem is that the dearth of genomic data in most cancer cases impedes processes required to develop cause and effect insights between genetics and treatment efficacy in the first place. Thus, without massive amounts of genetic data, there is no way to correlate genetic factors with treatment efficacy to develop justification for the expense associated with genetic testing in future cancer cases.

[0021] Yet one other problem posed by lack of genomic data is that if a researcher develops a genomic based treatment efficacy hypothesis based on a small genomic data set in a lab, the data needed to evaluate and clinically assess the hypothesis simply does not exist and it often takes months or even years to generate the data needed to properly evaluate the hypothesis. Here, if the hypothesis is wrong, the researcher may develop a different hypothesis which, again, may not be properly evaluated without developing a whole new set of genomic data for multiple patients over another several year period.

[0022] For some cancer states treatments and associated results are fully developed and understood and are generally consistent and acceptable (e.g., high cure rate, no long term effects, minimal or at least understood side effects, etc.). In other cases, however, treatment results cause and effect data associated with other cancer states is underdeveloped and/or inaccessible for several reasons. First, there are more than 250 known cancer types and each type may be in one of first through four stages where, in each stage, the cancer may have many different characteristics so that the number of possible “cancer varieties” is relatively large which makes the sheer volume of knowledge required to fully comprehend all treatment results unwieldy and effectively inaccessible.

[0023] Second, there are many factors that affect treatment efficacy including many different types of patient conditions where different conditions render some treatments more efficacious for one patient than other treatments or for one patient as opposed to other patients. Clearly capturing specific patient conditions or cancer state factors that do or may have a cause and effect relationship to treatment results is not easy and some causal conditions may not be appreciated and memorialized at all.

[0024] Third, for most cancer states, there are several different treatment options where each general option can be customized for a specific cancer state and patient condition set. The plethora of treatment and customization options in many cases makes it difficult to accurately capture treatment and results data in a normalized fashion as there are no clear standardized guidelines for how to capture that type of information.

[0025] Fourth, in most cases patient treatments and results are not published for general consumption and therefore are simply not accessible to be combined with other treatment and results data to provide a more fulsome overall data set. In this regard, many physicians see treatment results that are within an expected range of efficacy and conclude that those results cannot add to the overall cancer treatment knowledge base and therefore those results are never published. The problem here is that the expected range of efficacy can be large (e.g., 20% of patients fully heal and recover, 40% live for an extended duration, 40% live for an intermediate duration and 20% do not appreciably respond to a treatment plan) so that all treatment results are within an “expected” efficacy range and treatment result nuances are simply lost.

[0026] Fifth, currently there is no easy way to build on and supplement many existing illness-treatment-results databases so that as more data is generated, the new data and associated results cannot be added to existing databases as evidence of treatment efficacy or to challenge efficacy. Thus, for example, if a researcher publishes a study in a medical journal, there is no easy way for other physicians or researchers to supplement the data captured in the study. Without data supplementation over time, treatment and results corollaries cannot be tested and confirmed or challenged.

[0027] Sixth, the knowledge base around cancer treatments is always growing with different clinical trials in different stages around the world so that if a physician's knowledge is current today, her knowledge will be dated within months if not weeks. Thousands of oncological articles are published each year and many are verbose and/or intellectually arduous to consume (e.g., the articles are difficult to read and internalize), especially by extremely busy physicians that have limited time to absorb new materials and information. Distilling publications down to those that are pertinent to a specific physician's practice takes time and is an inexact endeavor in many cases.

[0028] Seventh, in most cases there is no clear incentive for physicians to memorialize a complete set of treatment and results data and, in fact, the time required to memorialize such data can operate as an impediment to collecting that data in a useful and complete form. To this end, prescribing and treating physicians are busy diagnosing and treating patients based on what they currently understand and painstakingly capturing a complete set of cancer state, treatment and results data without instantaneously reaping some benefit for patients being treated in return (e.g. a new insight, a better prescriptive treatment tool, etc.) is often perceived as a “waste” of time. In addition, because time is often of the essence in cancer treatment planning and plan implementation (e.g., starting treatment as soon as possible can increase efficacy in many cases), most physicians opt to take more time attending to their patients instead of generating perfect and fulsome treatments and results data sets.

[0029] Eighth, the field of next generation sequencing (“NGS”) for cancer genomics is new and NGS faces significant challenges in managing related sequencing, bioinformatics, variant calling, analysis, and reporting data. Next generation sequencing involves using specialized equipment such as a next generation gene sequencer, which is an automated instrument that determines the order of nucleotides in DNA and RNA. The instrument reports the sequences as a string of letters, called a read, which the analyst compares to one or more reference genomes of the same genes, which is like a library of normal and variant gene sequences associated with certain conditions. With no settled NGS standards, different NGS providers have different approaches for sequencing cancer patient genomics and, based on their sequencing approaches, generate different types and quantities of genomics data to share with physicians, researchers, and patients. Different genomic datasets exacerbate the task of discerning and, in some cases, render it impossible to discern, meaningful genetics-treatment efficacy insights as required data is not in a normalized form, was never captured or simply was never generated.

[0030] In addition to problems associated with collecting and memorializing treatment and results data sets, there are problems with digesting or consuming recorded data to generate useful conclusions. For instance, recorded cancer state, treatment and results data is often incomplete. In most cases physicians are not researchers and they do not follow clearly defined research techniques that enforce tracking of all aspects of cancer states, treatments and results and therefore data that is recorded is often missing key information such as, for instance, specific patient conditions that may be of current or future interest, reasons why a specific treatment was selected and other treatments were rejected, specific results, etc. In many cases where cause and effect relationships exist between cancer state factors and treatment results, if a physician fails to identify and record a causal factor, the results cannot be tied to existing cause and effect data sets and therefore simply cannot be consumed and added the overall cancer knowledge data set in a meaningful way.

[0031] Another impediment to digesting collected data is that physicians often capture cancer state, treatment and results data in forms that make it difficult if not impossible to process the collected information so that the data can be normalized and used with other data from similar patient treatments to identify more nuanced insights and to draw more robust conclusions. For instance, many physicians prefer to use pen and paper to track patient care and/or use personal shorthand or abbreviations for different cancer state descriptions, patient conditions, treatments, results and even conclusions. Using software to glean accurate information from hand written notes is difficult at best and the task is exacerbated when hand written records include personal abbreviations and shorthand representations of information that software simply cannot identify with the physician's intended meaning.

[0032] One positive development in the area of cancer treatment planning has been establishment of cancer committees or boards at cancer treating institutions where committee members routinely consider treatment planning for specific patient cancer states as a committee. To this end, it has been recognized that the task of prescribing optimized treatment plans for diagnosed cancer states is exacerbated by the fact that many physicians do not specialize in more than one or a small handful of cancer treatment options (e.g., radiation therapy, chemotherapy, surgery, etc.). For this reason, many physicians are not aware of many treatment options for specific ailment-patient condition combinations, related treatment efficacy and/or how to implement those treatment options. In the case of cancer boards, the idea is that different board members bring different treatment experiences, expertise and perspectives to bear so that each patient can benefit from the combined knowledge of all board members and so that each board member's awareness of treatment options continually expands.

[0033] While treatment boards are useful and facilitate at least some sharing of experiences among physicians and other healthcare providers, unfortunately treatment committees only consider small snapshots of treatment options and associated results based on personal knowledge of board members. In many cases boards are forced to extrapolate from “most similar” cancer states they are aware of to craft patient treatment plans instead of relying on a more fulsome collection of cancer state-treatment-results data, insights and conclusions. In many cases the combined knowledge of board members may not include one or several important perspectives or represent important experience bases so that a final treatment plan simply cannot be optimized.

[0034] To be useful cancer state, treatment and efficacy data and conclusions based thereon have to be rendered accessible to physicians, researchers and other interested parties. In the case of cancer treatments where cancer states, treatments, results and conclusions are extremely complicated and nuanced, physician and researcher interfaces have to present massive amounts of information and show many data corollaries and relationships. When massive amounts of information are presented via an interface, interfaces often become extremely complex and intimidating which can result in misunderstanding and underutilization. What is needed are well designed interfaces that make complex data sets simple to understand and digest. For instance, in the case of cancer states, treatments and results, it would be useful to provide interfaces that enable physicians to consider de-identified patient data for many patients where the data is specifically arranged to trigger important treatment and results insights. It would also be useful if interfaces had interactive aspects so that the physicians could use filters to access different treatment and results data sets, again, to trigger different insights, to explore anomalies in data sets, and to better think out treatment plans for their own specific patients.

[0035] In some cases specific cancers are extremely uncommon so that when they do occur, there is little if any data related to treatments previously administered and associated results. With no proven best or even somewhat efficacious treatment option to choose from, in many of these cases physicians turn to clinical trials.

[0036] Cancer research is progressing all the time at many hospitals and research institutions where clinical trials are always being performed to test new medications and treatment plans, each trial associated with one or a small subset of specific cancer states (e.g., cancer type, state, tumor location and tumor characteristics). A cancer patient without other effective treatment options can opt to participate in a clinical trial if the patient's cancer state meets trial requirements and if the trial is not yet fully subscribed (e.g., there is often a limit to the number of patients that can participate in a trial).

[0037] At any time there are several thousand clinical trials progressing around the world and identifying trial options for specific patients can be a daunting endeavor. Matching patient cancer state to a subset of ongoing trials is complicated and time consuming. Pairing down matching trials to a best match given location, patient and physician requirements and other factors exacerbates the task of considering trial participation. In addition, considering whether or not to recommend a clinical trial to a specific patient given the possibility of trial treatment efficacy where the treatments are by their very nature experimental, especially in light of specific patient conditions, is a daunting activity that most physicians do not take lightly. It would be advantageous to have a tool that could help physicians identify clinical trial options for specific patients with specific cancer states and to access information associated with trial options.

[0038] As described above, optimized cancer treatment deliberation and planning involves consideration of many different cancer state factors, treatment options and treatment results as well as activities performed by many different types of service providers including, for instance, physicians, radiologists, pathologists, lab technicians, etc. One cancer treatment consideration most physicians agree affects treatment efficacy is treatment timing where earlier treatment is almost always better. For this reason, there is always a tension between treatment planning speed and thoroughness where one or the other of speed and thoroughness suffers.

[0039] One other problem with current cancer treatment planning processes is that it is difficult to integrate new pertinent treatment factors, treatment efficacy data and insights into existing planning databases. In this regard, known treatment planning databases and application programs have been developed based on a predefined set of factors and insights and changing those databases and applications often requires a substantial effort on the part of a software engineer to accommodate and integrate the new factors or insights in a meaningful way where those factors and insights are properly considered along with other known factors and insights. In some cases the substantial effort required to integrate new factors and insights simply means that the new factors or insights will not be captured in the database or used to affect planning. In other cases the effort means that the new factors or insights are only added to the system at some delayed time after a software engineer has applied the required and substantial reprogramming effort. In still other cases, the required effort means that physicians that want to apply new insights and factors may attempt to do so based on their own experiences and understandings instead of in a more scripted and rules based manner. Unfortunately, rendering a new insight actionable in the case of cancer treatment is a literal matter of life and death and therefore any delay or inaccurate application can have the worst effect on current patient prognosis.

[0040] Another problem with existing cancer treatment efficacy databases and systems is that they are simply incapable of optimally supporting different types of system users. To this end, data access, views and interfaces needed for optimal use are often dependent upon what a system user is using the system for. For instance, physicians often want treatment options, results and efficacy data distilled down to simple correlations while a cancer researcher often requires much more detailed data access required to develop new hypothesis related to cancer state, treatment and efficacy relationships. In known systems, data access, views and interfaces are often developed with one consuming client in mind such as, for instance, physicians, pathologists, radiologists, a cancer treatment researcher, etc., and are therefore optimized for that specific system user type which means that the system is not optimized for other user types and cannot be easily changed to accommodate needs of those other user types.

[0041] With the advent of NGS it has become possible to accurately detect genetic alterations in relevant cancer genes in a single comprehensive assay with high sensitivity and specificity. However, the routine use of NGS testing in a clinical context faces several challenges. First, many tissue samples include minimal high quality DNA and RNA required for meaningful testing. In this regard, nearly all clinical specimens comprise formalin fixed paraffin embedded tissue (FFPET), which, in many cases, has been shown to include degraded DNA and RNA. Exacerbating matters, many samples available for testing contain limited amounts of tissue, which in turn limits the amount of nucleic acid attainable from the tissue. For this reason, accurate profiling in clinical specimens requires an extremely sensitive assay capable of detecting gene alterations in specimens with a low tumor percentage. Second, millions of bases within the tumor genome are assayed. For this reason, rigorous statistical and analytical approaches for validation are required in order to demonstrate the accuracy of NGS technology for use in clinical settings and in developing cause and effect efficacy insights.

[0042] Thus, what is needed is a system that is capable of efficiently capturing all treatment relevant data including cancer state factors, treatment decisions, treatment efficacy and exploratory factors (e.g., factors that may have a causal relationship to treatment efficacy) and structuring that data to optimally drive different system activities including memorialization of data and treatment decisions, database analytics and user applications and interfaces. In addition, the system should be highly and rapidly adaptable so that it can be modified to absorb new data types and new treatment and research insights as well as to enable development of new user applications and interfaces optimized to specific user activities.

BRIEF SUMMARY OF THE DISCLOSURE

[0043] It has been recognized that an architecture where system processes are compartmentalized into loosely coupled and distinct micro-services that consume defined subsets of system data to generate new data products for consumption by other micro-services as well as other system resources enables maximum system adaptability so that new data types as well as treatment and research insights can be rapidly accommodated. To this end, because micro-services operate independently of other system resources to perform defined processes where the only development constraints are related to system data consumed and data products generated, small autonomous teams of scientists and software engineers can develop new micro-services with minimal system constraints thereby enabling expedited service development.

[0044] The system enables rapid changes to existing micro-services as well as development of new micro-services to meet any data handling and analytical needs. For instance, in a case where a new record type is to be ingested into an existing system, a new record ingestion micro-service can be rapidly developed for new record intake purposes resulting in addition of the new record in a raw data form to a system database as well as a system alert notifying other system resources that the new record is available for consumption. Here, the intra-micro-service process is independent of all other system processes and therefore can be developed as efficiently and rapidly as possible to achieve the service specific goal. As an alternative, an existing record ingestion micro-service may be modified independent of other system processes to accommodate some aspect of the new record type. The micro-service architecture enables many service development teams to work independently to simultaneously develop many different micro-services so that many aspects of the overall system can be rapidly adapted and improved at the same time.

[0045] According to another aspect of the present disclosure, in at least some disclosed embodiments system data may be represented in several differently structured databases that are optimally designed for different purposes. To this end, it has been recognized that system data is used for many different purposes such as memorialization of original records or documents, for data progression memorialization and auditing, for internal system resource consumption to generate interim data products, for driving research and analytics, and for supporting user application programs and related interfaces, among others. It has also been recognized that a data structure that is optimal for one purpose often is sub-optimal for other purposes. For instance, data structured to optimize for database searching by a data scientist may have a completely different structure than data optimized to drive a physician's application program and associated user interface. As another instance, data optimized for database searching by a data scientist usually has a different structure than raw data represented in an original clinical medical record that is stored to memorialize the original record.

[0046] By storing system data in purpose specific data structures, a diverse array of system functionality is optimally enabled. Advantages include simpler and more rapid application and micro-service development, faster analytics and other system processes and more rapid user application program operations.

[0047] Particularly useful systems disclosed herein include three separate databases including a “data lake” database, a “data vault” database and a “data marts” database. The data lake database includes, among other data, original raw data as well as interim micro-service data products and is used primarily to memorialize original raw data and data progression for auditing purposes and to enable data recreation that is tied to prior points in time. The data vault database includes data structured optimally to support database access and manipulation and typically includes routinely accessed original data as well as derived data. The data marts database includes data structured to support specific user application programs and user interfaces including original as well as derived data.

[0048] One aspect of the present disclosure provides a computer system for evaluating a cancer treatment efficacy for a human subject that is undergoing, or has completed, a cancer treatment for a cancer. The computer system comprises one or more processors, and a memory. The memory stores instructions for performing a method using the one or more processors, the method comprises obtaining, from an electronic data store, first genomic sequencing data from cell free DNA drawn from a blood sample of the human subject. The first genomic sequencing data comprises a plurality of epigenetic patterns of cell free DNA within the blood sample. The method further comprises obtaining, from an electronic data store, second genomic sequencing data from cell free DNA drawn from the blood sample. The second genomic sequencing data comprises sequence data of cell free DNA within the blood sample. The method further comprises mapping the first genomic sequencing data to corresponding locations in a reference human genome. The method further comprises mapping the second genomic sequencing data to corresponding locations in the reference human genome. The method further comprises determining an absence or presence of at least a first genomic alteration, from among a plurality of predefined genomic alterations, in the blood sample based on at least (i) the second genomic sequencing data, and (ii) the locations in the reference human genome to which the second genomic sequencing data has been mapped. The method further comprises securely communicating, through a network connection, a structured data report that provides an assessment of the cancer treatment efficacy for the human subject responsive to at least the plurality of epigenetic patterns of cell free DNA within the blood sample and the absence or presence of at least the first genomic alteration in the blood sample.

[0049] In some embodiments, the method further comprises interfacing with a cloud service platform to perform at least a portion of the methods.

[0050] In some embodiments, the plurality of predefined genomic alterations consists of mutations in the genes listed in FIG. **22**.

[0051] In some embodiments, the plurality of predefined genomic alterations consists of mutations in the genes listed in FIG. **27**.

[0052] In some embodiments, the plurality of predefined genomic alterations consists of mutations in the genes listed in FIG. **28**.

[0053] In some embodiments, the first genomic alteration is a single nucleotide polymorphism in a gene.

[0054] In some embodiments, the first genomic alteration is an insertion or deletion mutation in a

gene.
[0055] In some embodiments, the first genomic alteration is a genomic rearrangement occurring in a gene.
[0056] In some embodiments, the method further comprises verifying that each genomic alteration in the plurality of predefined genomic alterations is a somatic genomic alteration.
[0057] In some embodiments, the assessment of the cancer treatment is in the form of a molecular residual disease status or a cancer recurrence status for the human subject, and the method further comprises determining, for each respective genomic alteration determined to be present, whether the respective genomic alteration is germline or somatic. In such embodiments, when the respective genomic alteration is determined to be germline, use of the respective genomic alteration is suppressed in the providing of the molecular residual disease (MRD) status or the cancer recurrence status for the human subject.
[0058] In some embodiments the method further comprises storing the structured data report in a subject data store.
[0059] In some embodiments, the method further comprises retrieving health data associated with the human subject, and augmenting the structured data report with the health data.
[0060] In some such embodiments, the health data comprises a cancer type associated with the human subject, and the augmenting the structured data report with the health data comprises including the cancer type in the data report.
[0061] In some such embodiments, the health data comprises details of a specimen collected from the human subject, and the augmenting the data report with the health data comprises including details of the specimen in the data report.
[0062] In some such embodiments, the health data comprises a diagnosis, a recurrence, a medication, a surgery, a response to treatment incurred by the human subject, an adverse effect to treatment incurred by the human subject, or an organoid modeling result, and the augmenting the data report comprises including the diagnosis, date of recurrence, the treatment incurred by the human subject, or an outcome of treatment in the data report.
[0063] In some such embodiments, the health data is retrieved from an electronic medical record or an electronic health record associated with the human subject.
[0064] In some such embodiments, the health data provides an indication of a status of a colorectal, ovarian, lung, or breast cancer for the human subject.
[0065] In some embodiments, the human subject is afflicted with colorectal, ovarian, lung, or breast cancer.
[0066] Another aspect of the present disclosure provides a non-transitory computer readable storage medium storing one or more programs, the one or more programs comprising instructions that, when executed by an electronic device with one or more processors and a memory, cause the electronic device to perform a method for evaluating a cancer treatment efficacy for a human subject that is undergoing, or has completed, a cancer treatment for a cancer, the method comprising and the of the methods disclosed herein.
[0067] To the accomplishment of the foregoing and related ends, the invention, then, comprises the features hereinafter fully described. The following description and the annexed drawings set forth in detail certain illustrative aspects of the invention. However, these aspects are indicative of but a few of the various ways in which the principles of the invention can be employed. Other aspects, advantages and novel features of the invention will become apparent from the following detailed description of the invention when considered in conjunction with the drawings.

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## Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0068] FIG. **1** is a schematic diagram illustrating a computer and communication system that is consistent with at least some aspects of the present disclosure:

[0069] FIG. **2** is a schematic diagram illustrating another view of the FIG. **1** system where functional components that are implemented by the FIG. **1** components are shown in some detail;

[0070] FIG. **3** is a schematic diagram illustrating yet another view of the FIG. **1** system where additional system components are illustrated;

[0071] FIG. **3***a* is a schematic diagram showing a data platform that is consistent with at least some aspects of the present disclosure;

[0072] FIG. **4** is a data handling flow chart that is consistent with at least some aspects of the present disclosure;

[0073] FIG. **5** is a flow chart that shows a process for ingesting raw data into the system and alerting other system components that the raw data is available for consumption;

[0074] FIG. **6** is a flow chart that shows a micro-service based process for retrieving data from a database, consuming that data to generate new data products and publishing the new data products back to a database while publishing an alert that the new data products are available for consumption;

[0075] FIG. **7** is a flow chart illustrating a process similar to the FIG. **6** process, albeit where the micro-service is an OCR service;

[0076] FIG. **8** is a is a flow chart illustrating a process similar to the FIG. **6** process, albeit where the micro-service is a data structuring service; and

[0077] FIG. **9** is a schematic view of an abstractor's display screen used to generate a structured data record from data in an unstructured or semi-structured record;

[0078] FIG. **10** is a schematic illustrating a multi-micro-service process for ingesting a clinical medical record into the system of FIG. **1**;

[0079] FIG. **11** is a schematic illustrating a multi-micro-service process for generating genomic sequencing and related data that is consistent with at least some aspects of the present disclosure;

[0080] FIG. **11***a* is a flow chart illustrating an exemplary variant calling process that is consistent with at least some aspects of the present disclosure;

[0081] FIG. **11***b* is a schematic illustrating an exemplary bioinformatics pipeline process that is consistent with at least some embodiments of the present disclosure;

[0082] FIG. **11***c* is a schematic illustrating various system features including a therapy matching engine;

[0083] FIG. **12** is a schematic illustrating a multi-micro-service process for generating organoid modelling data that is consistent with at least some aspects of the present disclosure;

[0084] FIG. **13** is a schematic illustrating a multi-micro-service process for generating a 3D model of a patient's tumor as well as identifying a large number of tumor features and characteristics that is consistent with at least some aspects of the present disclosure;

[0085] FIG. **14** is a screenshot illustrating a patient list view that may be accessed by a physician using the disclosed system to consider treatment options for a patient;

[0086] FIG. **15** is a screenshot illustrating an overview view that may be accessed by a physician using the disclosed system to review prior treatment or case activities related to the patient.

[0087] FIG. **16** is a screenshot illustrating screenshot illustrating a reports view that may be used to access patient reports generated by the system **100**;

[0088] FIG. **17** is a screenshot illustrating a second reports view that shows one report in a larger format;

[0089] FIG. **17***a* shows an initial view of an RNA sequence reporting screenshot that is consistent with at least some aspects of the present disclosure;

[0090] FIG. **18** is a screenshot illustrating an alterations view accessible by a physician to consider

molecular tumor alterations;
[0091] FIG. **18***a* is an exemplary top portion of a screenshot of a user interface for reporting and exploring approved therapies while FIG. **18***b* shows the lower portion of the FIG. **18***a* screenshot;
[0092] FIG. **19** is a screenshot illustrating a trials view in which a physician views information related to clinical trials on conjunction with considering treatment options for a patient;
[0093] FIG. **20** is a screenshot illustrating an immunotherapy screenshot accessible to a physician for considering immunotherapy efficacy options for treating a patient's cancer state;
[0094] FIG. **21** is a screenshot illustrating an efficacy exploration view where molecular differences between a patient's tumor and other tumors of the same general type are used a primary factor in generating the illustrated graph;
[0095] FIGS. **22***a* through **22***j* include an exemplary 1711 gene panel listing that may be interrogated during genomic sequencing in at least some embodiments of the present disclosure;
[0096] FIG. **23** includes a clinically actionable 130 gene panel listing that may be interrogated during genomic sequencing in at least some embodiments of the present disclosure;
[0097] FIG. **24** includes a clinically actionable 41 RNA based gene rearrangements listing that may be interrogated during genomic sequencing in at least some embodiments of the present disclosure;
[0098] FIG. **25** includes a table that lists exemplary variant data that is consistent with at least some aspects of the present disclosure;
[0099] FIG. **26** includes exemplary CVA data that is consistent with at least some implementations and aspects of the present disclosure;
[0100] FIGS. **27***a* through **27***d* includes additional gene panel tables that may be interrogated in at least some embodiments of the present disclosure;
[0101] FIGS. **28***a* and **28***b* include yet one other gene panel table that may be interrogated;
[0102] FIG. **29** is a bar chart illustrating data for a 500 patient group that clusters mutation similarities for gene, mutation type, and cancer type derived for an exemplary xT panel using techniques that are consistent with aspects of the present disclosure;
[0103] FIG. **30** is a bar chart comparing study results generated for the exemplary xT panel using at least some processes described in this specification with previously published pan-cancer analysis using an IMPACT panel;
[0104] FIG. **31** is a graph illustrating expression profiles for tumor types related to the exemplary xT panel described in the present disclosure;
[0105] FIG. **32** is a graph illustrating clustering of samples by TCGA cancer group in a t-SNE plot for the exemplary xT panel;
[0106] FIG. **33** is a plot of genomic rearrangements using DNA and RNA assays for the exemplary xT panel;
[0107] FIG. **34** is a schematic illustrating data related to one rearrangement detected via RNA sequencing related to the exemplary xT panel;
[0108] FIG. **35** is a schematic illustrating data related to a second rearrangement detected via RNA sequencing related to the exemplary xT panel;
[0109] FIG. **36** includes a chart that illustrates the distribution of TMB varied by cancer type identified using techniques that are consistent with at least some aspects of the present disclosure related to the exemplary xT panel;
[0110] FIG. **37** includes data represented on a two dimensional plot showing TMB on one axis and predicted antigenic mutations with RNA support on the other axis that was generated using techniques that are consistent with at least some aspects of the present disclosure related to the exemplary xT panel;
[0111] FIG. **38** includes additional data related to TMB generated using techniques that are consistent with at least some aspects of the present disclosure related to the exemplary xT panel;
[0112] FIG. **39** includes two schematics illustrating two gene expression scores for low and high TMB and MSI populations generated using techniques that are consistent with at least some

aspects of the present disclosure related to the exemplary xT panel;

[0113] FIG. **40** includes three schematics illustrating data related to propensity of different types inflammatory immune and non-inflammatory immune cells in low and high TMB samples generated for the related xT panel;

[0114] FIG. **41** includes a schematic illustrating data related to prevalence of CD274 expression in low and high TMB samples generated using techniques consistent with at least some aspects of the present disclosure generated for the related xT panel;

[0115] FIG. **42** includes two schematics illustrating correlations between CD274 expression and other cell types generated using techniques consistent with at least some aspects of the present disclosure generated for the related xT panel;

[0116] FIG. **43** is a schematic illustrating data generated via a 28 gene interferon gamma-related signature that is consistent with at least some aspects of the present disclosure;

[0117] FIG. **44** includes data shown as a graph illustrating levels of interferon gamma-related genes versus TMB-high, MSI-high and PDL1 IHC positive tumors generated using techniques consistent with at least some aspects of the present disclosure;

[0118] FIG. **45** includes a bar graph illustrating data related to therapeutic evidence as it varies among different cancer types generated using techniques consistent with at least some aspects of the present disclosure;

[0119] FIG. **46** includes a bar graph illustrating data related to specific therapeutic evidence matches based on copy number variants generating using techniques consistent with at least some aspects of the present disclosure;

[0120] FIG. **47** includes a bar graph illustrating data related to specific therapeutic evidence matches based on single nucleotide variants and indels generating using techniques consistent with at least some aspects of the present disclosure;

[0121] FIG. **48** includes a plot illustrating data related to single nucleotide variants and indels or CNVs by cancer type generating using techniques consistent with at least some aspects of the present disclosure;

[0122] FIG. **49** includes a bar graph illustrating data that shows percent of patients with gene calls and evidence for association between gene expression and drug response where the data was generated using techniques consistent with at least some aspects of the present disclosure;

[0123] FIG. **50** includes a bar graph illustrating response to therapeutic options based on evidence tiers and broken down by cancer type;

[0124] FIG. **51** includes a bar graph showing data related to patients that are potential candidates for immunotherapy broken down by cancer type where the data is based on techniques consistent with the present disclosure;

[0125] FIG. **52** is a bar graph presenting data related to relevant molecular insights for a patent group based on CNVs, indels, CNVs, gene expression calls and immunotherapy biomarker assays where the data was generated using techniques that are consistent with various aspects of the present disclosure;

[0126] FIG. **53** includes a bar graph illustrating disease-based trial matches and biomarker based match percentages based that reflect results of techniques that are consistent with at least some aspects of the present disclosure;

[0127] FIG. **54** includes a bar graph including data that shows exemplary distribution of expression calls by sample that was generated using techniques that are consistent with at least some aspects of the present disclosure;

[0128] FIG. **55** includes a bar graph including data that shows exemplary distribution of expression calls by gene that was generated using techniques that are consistent with at least some aspects of the present disclosure;

[0129] FIG. **56** includes a graph illustrating response evidence to therapies across all cancer types in an exemplary study using techniques consistent with at least some aspects of the present

disclosure;
[0130] FIG. **57** includes a graph illustrating evidence of resistance to therapies across all cancer types in an exemplary study using techniques consistent with at least some aspects of the present disclosure; and
[0131] FIG. **58** includes a graph illustrating therapeutic evidence tiers for all cancer types in an exemplary study using techniques consistent with at least some aspects of the present disclosure.
[0132] While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and are herein described in detail. It should be understood, however, that the description herein of specific embodiments is not intended to limit the invention to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.
DETAILED DESCRIPTION OF THE DISCLOSURE
[0133] The various aspects of the subject invention are now described with reference to the annexed drawings, wherein like reference numerals correspond to similar elements throughout the several views. It should be understood, however, that the drawings and detailed description hereafter relating thereto are not intended to limit the claimed subject matter to the particular form disclosed. Rather, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the claimed subject matter.
[0134] As used herein, the terms “component,” “system” and the like are intended to refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution. For example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a computer and the computer can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers or processors.
[0135] The word “exemplary” is used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs.
[0136] The phrase “Allelic Fraction” or “AF” will be used to refer to the percentage of reads supporting a candidate variant divided by a total number of reads covering a candidate locus.
[0137] The phrase “base pair” or “bp” will be used to refer to a unit consisting of two nucleobases bound to each other by hydrogen bonds. The size of an organism's genome is measured in base pairs because DNA is typically double stranded.
[0138] The phrase “Single Nucleotide Polymorphism” or “SNP” will be used to refer to a variation within a DNA sequence with respect to a known reference at a level of a single base pair of DNA.
[0139] The phrase “insertions and deletions” or “indels” will be used to refer to a variant resulting from the gain or loss of DNA base pairs within an analyzed region.
[0140] The phrase “Multiple Nucleotide Polymorphism” or “MNP” will be used to refer to a variation within a DNA sequence with respect to a known reference at a level of two or more base pairs of DNA, but not varying with respect to total count of base pairs. For example an AA to CC would be an MNP, but an AA to C would be a different form of variation (e.g., an indel).
[0141] The phrase “Copy Number Variation” or “CNV” will be used to refer to the process by which large structural changes in a genome associated with tumor aneuploidy and other dysregulated repair systems are detected. These processes are used to detect large scale insertions or deletions of entire genomic regions. CNV is defined as structural insertions or deletions greater than a certain base pair (“bp”) in size, such as 500 bp.
[0142] The phrase “Germline Variants” will be used to refer to genetic variants inherited from maternal and paternal DNA. Germline variants may be determined through a matched tumor-

normal calling pipeline.
[0143] The phrase “Somatic Variants” will be used to refer to variants arising as a result of dysregulated cellular processes associated with neoplastic cells. Somatic variants may be detected via subtraction from a matched normal sample.
[0144] The phrase “Gene Fusion” will be used to refer to the product of large scale chromosomal aberrations resulting in the creation of a chimeric protein. These expressed products can be non-functional, or they can be highly over or under active. This can cause deleterious effects in cancer such as hyper-proliferative or anti-apoptotic phenotypes.
[0145] The phrase “RNA Fusion Assay” will be used to refer to a fusion assay which uses RNA as the analytical substrate. These assays may analyze for expressed RNA transcripts with junctional breakpoints that do not map to canonical regions within a reference range. The term “Microsatellites” refers to short, repeated sequences of DNA.
[0146] The phrase “Microsatellite instability” or “MSI” refers to a change that occurs in the DNA of certain cells (such as tumor cells) in which the number of repeats of microsatellites is different than the number of repeats that was in the DNA when it was inherited. The cause of microsatellite instability may be a defect in the ability to repair mistakes made when DNA is copied in the cell.
[0147] “Microsatellite Instability-High” or “MSI-H” tumors are those tumors where the number of repeats of microsatellites in the cancer cell is significantly different than the number of repeats that are in the DNA of a benign cell. This phenotype may result from defective DNA mismatch repair. In MSI PCR testing, tumors where 2 or more of the 5 microsatellite markers on the Bethesda panel are unstable are considered MSI-H.
[0148] “Microsatellite Stable” or “MSS” tumors are tumors that have no functional defects in DNA mismatch repair and have no significant differences in microsatellite regions between tumor and normal tissue.
[0149] “Microsatellite Equivocal” or “MSE” tumors are tumors with an intermediate phenotype that cannot be clearly classified as MSI-H or MSS based on the statistical cutoffs used to define those two categories.
[0150] The phrase “Limit of Detection” or “LOD” refers to the minimal quantity of variant present that an assay can reliably detect. All measures of precision and recall are with respect to the assay LOD.
[0151] The phrase “BAM File” means a (B) inary file containing (A) lignment (M) aps that include genomic data aligned to a reference genome.
[0152] The phrase “Sensitivity of called variants” refers to a number of correctly called variants divided by a total number of loci that are positive for variation within a sample.
[0153] The phrase “specificity of called variants” refers to a number of true negative sites called as negative by an assay divided by a total number of true negative sites within a sample. Specificity can be expressed as (True negatives)/(True negatives+false positives).
[0154] The phrase “Positive Predictive Value” or “PPV” means the likelihood that a variant is properly called given that a variant has been called by an assay. PPV can be expressed as (number of true positives)/(number of false positives+number of true positives).
[0155] The disclosed subject matter may be implemented as a system, method, apparatus, or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer or processor based device to implement aspects detailed herein. The term “article of manufacture” (or alternatively, “computer program product”) as used herein is intended to encompass a computer program accessible from any computer-readable device, carrier, or media. For example, computer readable media can include but are not limited to magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips . . . ), optical disks (e.g., compact disk (CD), digital versatile disk (DVD) . . . ), smart cards, and flash memory devices (e.g., card, stick). Additionally it should be appreciated that a carrier wave can be employed to carry computer-readable electronic data such as those used in transmitting and

receiving electronic mail or in accessing a network such as the Internet or a local area network (LAN). Of course, those skilled in the art will recognize many modifications may be made to this configuration without departing from the scope or spirit of the claimed subject matter.

[0156] Unless indicated otherwise, while the disclosed system is used for many different purposes (e.g., data collection, data analysis, treatment, research, etc.), in the interest of simplicity and consistency, the overall disclosed system will be referred to hereinafter as “the disclosed system”.

I. System Overview

[0157] Referring now to the figures that accompany this written description and more specifically referring to FIG. **1**, the present disclosure will be described in the context of an exemplary system **100** where data is received at a system server **150** from many different data sources **102**, is stored in a database **160**, is manipulated in many different ways by internal system micro-service programs to condition or “shape” the data to generate new interim data or to structure data in different structured formats for consumption by user application programs and to then drive the user application programs to provide user interfaces via any of several different types of user interface devices. While a single server **150** and a single database **160** are shown in FIG. **1** in the interest of simplifying this explanation, it should be appreciated that in most cases, the system **100** will include a plurality of distributed servers and databases that are linked via local and/or wide area networks and/or the Internet or some other type of communication infrastructure. An exemplary simplified communication network is labelled **80** in FIG. **1**. Network connections can be any type including hard wired, wireless, etc., and may operate pursuant to any suitable communication protocols.

[0158] The disclosed system **10** enables many different system clients to securely link to server **150** using various types of computing devices to access system application program interfaces optimized to facilitate specific activities performed by those clients. For instance, in FIG. **1** a physician **10** is shown using a laptop computer (not labelled) to link to server **150**, an abstractor specialist **20** is shown using a tablet type computing device to link, another specialist **30** is shown using a smartphone device to link to server **150**, etc. Other types of personal computing devices are contemplated including virtual and augmented reality headsets, projectors, wearable devices (e.g., a smart watch, etc.). FIG. **1** shows other exemplary system users linked to server **150** including a partner researcher **40**, a provider researcher **50** and a data sales specialist **60**, all of which are shown using laptop computers.

[0159] In at least some embodiments when a physician uses system **100**, a physician's user interface(s) is optimally designed to support typical physician activities that the system supports including activities geared toward patient treatment planning. Similarly, when a researcher like a pathologist or a radiologist uses system **100**, interfaces optimally designed to support activities performed by those system clients are provided.

[0160] System specialists (e.g. employees of the provider that controls/maintains overall system **100**) also use interface computing devices to link to server **150** to perform various processes and functions. In FIG. **1** exemplary system specialists include abstractor **20**, the dataset sales specialist **60** and a “general” specialist **30** referred to as a “lab, modeling, radiology” specialist to indicate that the system accommodates many different additional specialist types. Different specialists will use system **100** to perform many different functions where each specialist requires specific skill sets needed to perform those functions. For instance, abstractor specialists are trained to ingest clinical records from sources **102** and convert that data to normalized and system optimized structured data sets. A lab specialist is trained to acquire and process non-tumorous patient and/or tumor tissue samples, grow organoids, generate one or both of DNA and RNA genomic data for one or each of non-tumorous and tumorous tissue, treat organoids and generate results. Other specialists are trained to assess treatment efficacy, perform data research to identify new insights of various types and/or to modify the existing system to adapt to new insights, new data types, etc. The system interfaces and tool sets available to provider specialists are optimized for specific needs and tasks performed by those specialists.

[0161] Referring yet again to FIG. **1**, system database **160** includes several different sub-databases including, in at least some embodiments, a data lake database **170** (hereinafter "the lake database"), a data vault database **180**, a data marts database **190** and a system services/applications and integration resource database **195**. While database **195** is shown to includes several different types of information as well as system programs, in other cases one or each of the sets of information or programs in database **195** may be stored in a different one of the databases **170**, **180** or **190**. In general, data lake database **170** is used to store several different data types including system reference data **162**, system administration data **164**, infrastructure data **166**, raw source data **168** and micro-service data products **172** (e.g., data generated by micro-services).

[0162] Reference data **162** includes references and terminology used within data received from source devices **102** when available such as, for instance, clinical code sets, specialized terms and phrases, etc. In addition, reference data **162** includes reference information related to clinical trials including detailed trial descriptions, qualifications, requirements, caveats, current phases, interim results, conclusions, insights, hypothesis, etc.

[0163] In at least some cases reference data **162** includes gene descriptions, variant descriptions, etc. Variant descriptions may be incorporated in whole or in part from known sources, such as the Catalogue of Somatic Mutations in Cancer (COSMIC) (Wellcome Sanger Institute, operated by Genome Research Limited, London, England, available at https://cancer.sanger.ac.uk/cosmic). In some cases, reference data **162** may structure and format data to support clinical workflows, for instance in the areas of variant assessment and therapies selection. The reference data **162** may also provide a set of assertions about genes in cancer and evidence-based precision therapy options. Inputs to reference data **162** may include NCCN, FDA, PubMed, conference abstracts, journal articles, etc. Information in the reference data **162** may be annotated by gene; mutation type (somatic, germline, copy number variant, fusion, expression, epigenetic, somatic genome wide, etc.); disease; evidence type (therapeutic, prognostic, diagnostic, associated, etc.); and other notes.

[0164] Referring still to FIG. **1**, reference data **162** may further comprise gene curation information. A sequencing panel often has a predetermined number of gene profiles that are sequenced as part of the panel. For instance, one type of sequencing panel in the market (i.e., xT, Tempus Labs, Inc, Chicago, Illinois) makes use of **595** gene profiles (see tables in FIG. **27** series of figures) while another makes use of **1711** gene profiles (see tables in FIG. **22** series of figures). Reference data **162** may store a centralized gene knowledge base and comprise variant prioritization and filtering information that may be utilized for Gain Of Function (GOF), Loss Of Function (LOF), CNV, and fusions. For purposes of precision care, evidence may be annotated based on mutation type and disease; therapeutic evidence may include drug(s) and effect (response, resistance, etc.); prognostic effect may include outcome (favorable, unfavorable, etc.). Therapeutic evidence and prognostic evidence may include evidence source level (preclinical, case study, clinical research, guidelines, etc.). Preclinical information may be from mouse models, PDX, cell lines, etc. Case study information may be from groups of one or more patients. Clinical research may be information from a larger study or results from clinical trials. Guideline information may come from NCCN, WHO, etc.

[0165] The administrative data **164** includes patient demographic data as well as system user information including user identifications, user verification information (e.g., usernames, passwords, etc.), constraints on system features usable by specific system users, constraints on data access by users including limitations to specific patient data, data types, data uses, time and other data access limits, etc.

[0166] In at least some cases system **100** is designed to memorialize entire life cycles of every dataset or element collected or generated by system **100** so that a system user can recreate any dataset corresponding to any point in time by replicating system processes up to that point in time. Here, the idea is that a researcher or other system user can use this data re-creation capability to verify data and conclusions based thereon, to manipulate interim data products as part of an exploration process designed to test other hypothesis based on system data, etc. To this end, infrastructure data **166** includes complete data storage, access, audit and manipulation logs that can be used to recreate any system data previously generated. In addition, infrastructure data **166** is usable to trace user access and storage for access auditing purposes.

[0167] Referring still to FIG. **1**, lake database **170** also includes raw unmodified data **168** from sources **102**. For instance, original clinical medical records from physicians are stored in their original format as are any medical images and radiology reports, pathology reports, organoid documentation, and any other data type related to patient treatment, treatment efficacy, etc. In addition the raw original data, metadata related thereto is also identified and stored at **168**. Exemplary metadata includes source identity, data type, date and time data received, any data formatting information available, etc. The metadata listed here is not exhaustive and other metadata types may also be obtained and stored. Raw sequencing data, such as BAM files, may be stored in lake database **170**. Unless indicated otherwise hereafter, the data stored in lake database **170** will be referred to generally as “lake data”.

[0168] It has been recognized that a fulsome database suitable for cancer research and treatment planning must account for a massive number of complex factors. It has also been recognized that the unstructured or semi-structured lake data is unsuitable for performing many data search processes, analytics and other calculations and data manipulations that are required to support the overall system. In this regard, searching or otherwise manipulating a massive database data set that includes data having many disparate data formats or structures can slow down or even halt system applications. For this reason the disclosed system converts much of the lake data to a system data structure optimized for database manipulation (e.g., for searching, analyzing, calculating, etc.). For example, genomic data may be converted to JSON or Apache Parquet format, however, others are contemplated. The optimized structured data is referred to herein as the “data vault database” **180**.

[0169] Thus, in FIG. **1**, data vault database **180** includes data that has been normalized and optimally structured for storage and database manipulation. For instance, raw original clinical medical records stored at **168** in lake database **170** may be processed to normalize data formats and placed in specific structured data fields optimized for data searching and other data manipulation processes. For instance, raw original clinical medical records, such as progress notes, pathology reports, etc. may be processed into specific structured data fields. Structured data fields may be focused in certain clinical areas, such as demographics, diagnosis, treatment and outcomes, and genetic testing/labs. For instance, structured diagnosis information may include primary diagnosis; tissue of origin; date of diagnosis; date of recurrence; date of biochemical recurrence; date of CRPC; alternative grade; gleason score; gleason score primary; gleason score secondary; gleason score overall; lymphovascular invasion; perineural invasion; venous invasion. Structured diagnosis information may also include tumor characterization, which may be described with a set of structured data, including the type of characterization; date of characterization; diagnosis; standard grade; AJCC values such as AJCC status, AJCC status T, AJCC status N, AJCC Status M, AJCC status stage, and FIGO status stage. Structured diagnosis information may also include tumor size, which may be described with a set of structured size data, including tumor size (greatest dimension), tumor size measure, and tumor size units. Structured diagnosis information may also include structured metastases information. Each metastasis may be described with a set of structured data, including location, date of identification, tumor size, diagnosis, grade, and AJCC values. Structured diagnosis information may also include additional diagnoses. Additional diagnoses may be described with a set of structured data, including tissue of origin, date of diagnosis, date of recurrence, date of biochemical recurrence, date of CRPC, tumor characterizations, and metastases.

[0170] As another instance, 2 dimensional slice type images through a patient's tumor may be used to generate a normalized 3 dimensional radiological tumor model having specific attributes of

interest and those attributes may be gleaned and stored along with the 3D tumor model in the structured data vault for access by other system resources. In FIG. **2**, the data vault database **180** is shown including a structured clinical database **181** for storage of structured clinical data, a molecular sequencing database **183** for storage of molecular sequencing data, a structure imaging database **185** for storage of imaging data, and a predictive modeling database **187** for storage of organoid and other modeling data. Additional databases for specific lines of data may also be added to the data vault database **180**. RNA sequencing data in the molecular sequencing data may be normalized, for instance using the methods disclosed in U.S. Provisional Patent App. No. 62/735,349, METHODS OF NORMALIZING AND CORRECTING RNA EXPRESSION DATA, incorporated by reference herein in its entirety. Unless indicated otherwise hereafter, the phrase “canonical data” will be used to refer to the data vault data in its system optimized structured form.
[0171] It has further been recognized that certain data manipulations, calculations, aggregates, etc., are routinely consumed by application programs and other system consumers on a recurring albeit often random basis. By shaping at least subsets of normalized system data, smaller sub-databases including application and research specific data sets can be generated and published for consumption by many different applications and research entities which ultimately speeds up the data access and manipulation processes.
[0172] Thus, in FIG. **1**, data marts database **190** includes data that is specifically structured to support user application programs **194** and/or specific research activities **196**. Here, it is contemplated that different user application programs may require different data models (e.g., different data structures) and therefore data marts **190** will typically include many different application or research specific structured data sets. For instance, a first data mart data set may include data arranged consistent with a first data structure model optimized to support a physician's user interfaces, a second data mart data set may include data arranged consistent with a second data structure model optimized to support a radiologist specialist, a third data mart data set may include data arranged consistent with a third data structure model optimized to support a partner researcher, and so on. A single user type may have multiple data mart data sets structured to support different workflows on the same or different raw data.
[0173] Similarly, in the case of specific research activities, specific data sets and formats are optimal for specific research activities and the data marts provide a vehicle by which optimized data sets are optimally structured to ensure speedy access and manipulation during research activities. Unless indicated otherwise hereafter, the phrase “mart data” will be used to generally refer to data stored in the data marts **190**.
[0174] In most cases mart data is mined out of the data vault **180** and is restructured pursuant to application and research data models to generate the mart data for application and research support. In some embodiments system orchestration modules or software programs that are described hereafter will be provided for orchestrating data mining in the system databases as well as restructuring data per different system models when required.
[0175] Referring still to FIG. **1**, the system services/applications/integration resources database **195** includes various programs and services run by system server **150** to perform and/or guide system functions. To this end, exemplary database **195** includes system orchestration modules/resources **184**, a set of first through N micro-services collectively identified by numeral **186**, operational user application programs **188** and analytical user application programs **192**.
[0176] Orchestration modules/resources **184** include overall scheduling programs that define workflows and overall system flow. For instance, one orchestration program may specify that once a new unstructured or semi-structured clinical medical record is stored in lake database **170**, several additional processes occur, some in series and some in parallel, to shape and structure new data and data derived from the new data to instantiate new sets of canonical data and mart data in databases **180** and **190**. Here, the orchestration program would manage all sub-processes and data handoffs required to orchestrate the overall system processes. One type of orchestration program that could

be utilized is a programmatic workflow application, which uses programming to author, schedule and monitor "workflows". A "workflow" is a series of tasks automatically executed in whole or in part by one or more micro-services. In one embodiment, the workflow may be implemented as a series of directed acyclic graphs (DAGs) of tasks or micro-services.

[0177] Micro-services **186** are system services that generate interim system data products to be consumed by other system consumers (e.g., applications, other micro-services, etc.). In FIG. **1**, first through Nth micro-service data products corresponding to micro-services **186** are shown stored in lake database **170** at **172**. When a micro-service data product is published to lake database **170**, a data alert or event is added to a data alerts list **169** to announce availability of the newly published data for consumption by other micro-services, application programs, etc. Micro-services are independent and autonomous in that, once a service obtains data required to initiate the service, the service operates independent of other system resources to generate output data products.

[0178] In many cases micro-services are completely automated software programs that consume system data and generate interim data products without requiring any user input. For instance, an exemplary fully automated micro-service may include an optical character recognition (OCR) program that accesses an original clinical record in the raw source data **168** and performs an OCR process on that data to generate an OCR tagged clinical record which is stored in lake database **170** as a data product **172**. As another instance, another fully automated micro-service may glean data subsets from an OCR tagged clinical record and populate structured record fields automatically with the gleaned data as a first attempt to convert unstructured or semi-structured raw data to a system optimized structure.

[0179] In other cases a micro-service requires at least some system user activities including, for instance, data abstraction and structuring services or lab activities, to generate interim data products **172**. For instance, in the case of clinical medical record ingestion, in many cases an original clinical record will be unstructured or semi-structured and structuring will require an abstractor specialist **20** (see again FIG. **1**) to at least verify data in structured data record fields and in many cases to manually add data to those fields to generate a completely instantiated instance of the structured record as a data product **172**. As another instance, in the case of genetic sequencing, a lab technician is required to obtain and load sample tumor or other tissue into a sequencing machine as part of a sequencing process. In cases where a service requires at least some user activities, the service will typically be divided into separate micro-services where a user application operates on a micro-service data product to queue user activities in a user work queue or the like and a separate micro-service responds to the user activity being completed to continue an overall process. While this disclosure describes a small set of micro-services, a working system **100** will typically employ a massive number (e.g., hundreds or even many thousands) of micro-services to drive all of the system capabilities contemplated. It is possible that in the life cycle of analysis for a patient that hundreds or thousands of executions of micro-services will be performed.

[0180] In an embodiment, a micro-service creates a data product that may be accessed by an application, where the application provides a worklist and user interface that allows a user to act upon the data product. One example set of micro-services is the set of micro-services for genomic variant characterization and classification. An exemplary micro-service set for genomic variant characterization includes but is not limited to the following set: (1) Variant characterization (a data package containing characterized variant calls for a case, which may include overall classification, reference criteria and other singles used to determine classification, exclusion rules, other flags, etc.); (2) Therapy match (including therapies matched to a variant characterization's list of SNV, indel, CNV, etc. variants via therapy templates); (3) Report (a machine-readable version of the data delivered to a physician for a case); (4) Variants reference sets (a set of unique variants analyzed across all cases); (5) Unique indel regions reference sets (gene-specific regions where pathogenic inframe indels and/or frameshift variants are known to occur); (6) DNA reports; (7) RNA reports; (8) Tumor Mutation Burden (TMB) calculations, etc. Once genomic variant characterization and classification has been completed, other applications and micro-services provide tools for variant scientists or other clinicians or even other micro-services to act upon the data results.

[0181] Referring still to FIG. **1**, each micro-service includes a service specification including definitions of data that the specified service is to consume, micro-service code defining the service to be performed by the specific micro-service and a definition of the data that is to be published to the lake as an interim data product **172**. In each case, the service to be performed includes monitoring the data alerts list **169** or published data on the system communication network for data to be consumed (e.g., monitor for data that fits subscriptions associated with the microservice) by the service and, once the service generates a data product, publishing that data product to the data lake and placing an alert in alerts list **169** or publishing that data. In operation, when a micro-service is to consume a published data product, the service obtains the data product, consumes the product as part of performing the service, publishes new data product(s) to lake database **170** and then places a new data alert in list **169** to announce to other system consumers that the new data is ready for consumption.

[0182] Another system for asynchronous communication between micro-services is a publish-subscribe message passing ("pub/sub") system which uses the alerts list **169**. In this system type, alerts list **169** may be implemented in the form of a message bus. One example of a message bus that may be utilized is Amazon Simple Notifications Service (SNS). In this system type, micro-services publish messages about their activities on message bus topics that they define. Other micro-services subscribe to these messages as needed to take action in response to activities that occur in other micro-services.

[0183] In at least some embodiments, micro-services are not required to directly subscribe to SNS topics. Rather, they set up message queues via a queue service, and subscribe their queues to the SNS Topics that they are interested in. The micro-services then pull messages from their queues at any time for processing, without worrying about missing messages. One example of a queue service is the Amazon Simple Queue Service (SQS) although others are contemplated.

[0184] Granularity of SNS topics may be defined on a message subject basis (for instance, 1 topic per message subject), on a domain object basis (for instance, one topic per domain object basis), and/or on a per micro-service basis (for instance, one topic per micro-service basis). Message content may include only essential information for the message in order to prioritize small message size. In at least some cases message content is architectured to avoid inclusion of patient health information or other information for which authorization is required to access.

[0185] Different alerts may be employed throughout the system. For instance, alerts may be utilized in connection with the registration of a patient. One example of an alert is "services-patients.created", which is triggered by creation of a new patient in the system. Alerts may be utilized in connection with the analysis of variant call files. One example is "variant-analysis_staging", which is triggered upon the completion of a new variant calling result. Another example is "variant-analysis_staging.ready", which is triggered upon completed ingestion of all input files for a variant calling result. Another example is "case_staging.ready", which is triggered when information in the system is ready for manual user review. Many other alerts are contemplated.

[0186] Both orchestration workflows and micro-service alerts may be employed in the system, either alone or in combination. In an example, an event-based micro-service architecture may be utilized to implement a complex workflow orchestration. Orchestrations may be integrated into the system so that they are tailored for specific needs of users. For instance, a provider or another partner who requires the ability to provide structured data into the lake may utilize a partner-specific orchestration to land structured data in the lake, pre-process files, map data, and load data into the data fault. As another example, a provider or other partner who requires the ability to provide unstructured data into the lake may utilize a partner-specific orchestration for pre-processing and providing unstructured data to the data lake. As another example, an orchestration may, upon publishing of data that is qualified for a particular use case (such as for research, or third-party delivery), transform the data and load it into a columnar data store technology. As another example, a "data vault to clinical mart" orchestration may take stable points in time of the data published to data vault by other orchestrations; transform the data into a mart model, and transform the mart data through a de-identification pipeline. As another example, a "commercial partner egress file gateway" may utilize a cohort of patients whose data is defined for delivery, sourcing the data from de-identified data marts and the data lake (including molecular sequencing data) and publish the same to a third-party partner.

[0187] Referring still to FIG. **1**, operational and analytical applications **188** and **192**, respectively, are application programs that provide functionality to various system user types as well as interfaces optimized for use by those system users. Operational applications **188** include application programs that are primarily required to enable cancer state treatment planning processes for specific patients. For instance, operational applications include application programs used by a cancer treating physician to assess treatment options and efficacy for a specific patient. As another instance, operational applications also include application programs used by an abstractor specialist to convert unstructured raw clinical medical records or semi-structured records to system optimized structured records. As another instance, operational applications may also include application programs used by bioinformatics scientists or molecular pathologists to annotate variants. As another instance, operational applications also include application programs used by clinicians to determine whether a patient is a good match for a clinical trial. As yet one other instance, operational applications may include application programs used by physicians to finalize patient reports.

[0188] Analytical applications **192**, in contrast, include application programs that are provided primarily for research purposes and use by either provider client researchers or provider specialist researchers. For instance, analytical applications **192** include programs that enable a researcher to generate and analyze data sets or derived data sets corresponding to a researcher specified subset of de-identified (e.g., not associated with a specific patient) cancer state characteristics. Here, analysis may include various data views and manipulation tools which are optimized for the types of data presented. Some applications may have features of both analytical applications **192** and operational applications **188**.

II. System Database Architecture And General Data Flow

[0189] Referring now to FIG. **2**, a second representation of disclosed system **100** shows many of the components shown in FIG. **1** in an operational arrangement. The FIG. **2** system includes system data sources **102** and operational system components including an integration layer **220** in addition to the lake database **170**, data vault database **180**, operational applications **188** and analytical applications **192** that are described above. Exemplary data sources **102** include physician clinical records systems **200**, radiology imaging systems **202**, provider genomic sequencers **204**, organoid modeling labs **206**, partner genomic sequencers **208** and research partner records systems **210**. The source data types are only exemplary and are not intended to be limiting. In fact, it is contemplated that many other data source types generating other clinically relevant data types will be added to the system over time as other sources and data types of interest are identified and integrated into the overall system.

[0190] Referring again to FIG. **2**, integration layer **220** includes integration gateways **312**/**314**, a data lake catalog **226** and the data marts database **190** described above with respect to FIG. **1**. The integration gateways receive data files and messages from sources **102**, glean metadata from those files and messages and route those files and messages on to other system components including data lake database **170** and catalog **226** as well as various system applications. New files are stored in lake database **170** and metadata useful for searching and otherwise accessing the lake data is stored in catalog **226**. Again, non-structured and semi-structured raw and micro-service data is stored in lake database **170** and system optimized structured data is stored in vault database **180** while application optimized structured data is stored in data marts database **190**.

[0191] Referring again to FIG. **2**, system users **10**, **20**, **30** **40**, **50** and **60** access system data and functionality via the operational and/or analytical applications **188** and **192**, respectively. In some instances, in order to protect patient confidentiality, the system user cannot have access to patient medical records that are tied to specific and identified patients. For this reason, integration layer **220** may include a de-identification module which accesses system data, scrubs that data to remove any specific patient identification information and then serves up the de-identified data to the application platform. In other examples, the data vault database may have its structure duplicated, such that a de-identified copy of the data in the data vault database **180** is retained separately from the non de-identified copy of the data in the data vault database. Data in the de-identified copy may be stripped of its identifiers, including patient names; geographic subdivisions smaller than a state, including street address, city, county, precinct, ZIP code, and their equivalent geocodes, except for the initial three digits of the ZIP code if, according to the current publicly available data from the Bureau of the Census: (1) The geographic unit formed by combining all ZIP codes with the same three initial digits contains more than 20,000 people; and (2) The initial three digits of a ZIP code for all such geographic units containing 20,000 or fewer people is changed to 000; elements of dates (except year) for dates that are directly related to an individual, including birth date, admission date, discharge date, death date, and all ages over 89 and all elements of dates (including year) indicative of such age, except that such ages and elements may be aggregated into a single category of age 90 or older; Telephone numbers; Vehicle identifiers and serial numbers, including license plate numbers; Fax numbers; Device identifiers and serial numbers; Email addresses; Web Universal Resource Locators (URLs); Social security numbers; Internet Protocol (IP) addresses; Medical record numbers; Biometric identifiers, including finger and voice prints; Health plan beneficiary numbers; Full-face photographs and any comparable images; Account numbers and other unique identifying numbers, characteristics, or codes; and Certificate/license numbers. Because data in the data vault database **180** is structured, much of the information not permitted for inclusion in the de-identified copy is absent by virtue of the fact that a structured location does not exist for inclusion of such information. For instance, the structure of the data vault database for storing the de-identified copy may not include a field for storing a social security number. As another example, data in the data vault database may be segregated by customer. For example, if one physician **10** wishes for his or her patients to have their data segregated from other data in the data lake database **170**, their data may be segregated in a single tenant data vault, such as the single tenant data vault arrangement shown in FIG. **3***a*.

[0192] Many users employing the operational applications **188** do have physician-patient relationships, or otherwise are permitted to access records in furtherance of treatment, and so have authority to access patent identified medical, healthcare and other personal records. Other users employing the operational applications have authority to access such records as business associates of a health care provider that is a covered entity. Therefore, in at least some cases, operational applications will link directly into the integration layer of the system without passing through de-identification module **224**, or will provide access to the non de-identified data in the database **160**. Thus, for instance, a physician treating a specific patient clearly requires access to patient specific information and therefore would use an operational application that presents, among other information, patient identifying information.

[0193] In some cases, users employing operational applications will want access to at least some de-identified analytical applications and functionality. For instance, in some cases an operational application may enable a physician to compare a specific patient's cancer state to multiple other patient's cancer states, treatments and treatment efficacies. Here, while the physician clearly needs access to her patient's identifying information and state factors, there is no need and no right for the physician to have access to information specifically identifying the other patients that are associated with the data to be compared. Thus, in some cases one operational application will access a set of patient identified data and other sets of patient de-identified data and may consume all of those data sets.

[0194] Referring now to FIG. **3**, a system representation **100** akin to the one in FIG. **2** is shown, albeit where the FIG. **3** representation is more detailed. In FIG. **3** integration layer **220** includes separate message and file gateways **312** and **314**, respectively, an event reporting bus **316**, system micro-services **186**, various data lake APIs **332**, **334** and **336**, an ETL module **338**, data lake query and analytics modules **346** and **348**, respectively, an ETL platform **360** as well as data marts database **190**.

[0195] Referring to FIG. **3**, sources **102** are linked via the internet or some other communication network to system **100** via message gateway **312** and file gateway **314**. Messages received from data sources **102** at gateway **312** are forwarded on to event bus **326** which routes those messages to other system modules as shown. Messages from other system modules can be routed to the data sources via message gateway **312**.

[0196] File gateway **314** receives source files and controls the process of adding those files to lake database **170**. To this end, the file gateway runs system access security software to glean metadata from any received file and to then determine if the file should be added to the lake database **170** or rejected as, for instance, from an unauthorized source. Once a file is to be added to the lake database, gateway **314** transfers the file to lake database **170** for storage, uses the metadata gleaned from the file to catalog the new file in the lake catalog **226** and posts an alert in the data alert list **169** (see again FIG. **1**) announcing that the new data has been published to the lake for consumption.

[0197] Referring still to FIG. **3**, a subset of micro-services monitoring alert list **169** for data of the type published to lake database **170** access the new data or consume that data when published to the network, perform their data consumption processes, publish new data products to lake database **170** and post new data alerts in list **169** or publish the new data on the network per the publication-subscription architecture described above. In cases where system user activities are required as part of a micro-service, the service schedules those activities to be completed by provider specialists when needed and ingests data generated thereby, eventually publishing new data products to the lake database **170**.

[0198] The orchestration modules and resources monitor the entire data process and determine when data lake data is to be replicated within the data vault and/or within the data marts in different system or application optimized model formats. Whenever lake data is to be restructured and placed in the data vault or the data marts, ETL platform **360** extracts the data to restructure, transforms the data to the system or application specific data structure required and then loads that data into the respective database **180** or **190**. In some cases it is contemplated that ETL platform may only be capable of transforming data from the data lake structure to the data vault structure and from the data vault structure to the application specific data models required in data marts **190**.

[0199] Referring still to FIG. **3**, analytical applications **192** are shown to include, among other applications, “self-service” applications. Here, the phrase “self-service” is used to refer to applications that enable a system user to, in effect, use query tools and data visualization tools, to access and manipulate data sets that are not optimally supported by other user applications. Here, the idea is that, especially in the context of research, system users should not be constrained to specific data sets and analysis and instead should be able to explore different data sets associated with different cancer state factors, different treatments and different treatment efficacies. The self-service tools are designed to allow an authorized system user to develop different data visualizations, unique SQL or other database queries and/or to prepare data in whatever format desired. Hereinafter, unless indicated otherwise, the term “explore” will be used to refer to any self-service activities performed within the disclosed system.

[0200] Referring still to FIG. **3**, self-service applications **356** enable a system user to explore all system databases in at least some embodiments including the data marts **190**, the lake database **170** and the data vault database **180**. In other embodiments, because lake database **170** data is either unstructured or only semi-structured, self-service applications may be limited to exploring only the data mart database **190** or the data vault database **180**.

III. Data Ingestion, Normalization And Publication

[0201] Referring to FIG. **4**, a high level data distribution process **400** is illustrated that is consistent with at least some aspects of the present disclosure. At process block **402**, data is collected from various data sources **102** (see again FIGS. **1** through **3**) and at block **404**, assuming that data is to be ingested into the system **100**, the data is stored in lake database **170**. Here, data collection is continual over time as more and more data for increasing the system knowledge base is generated regularly by physicians, provider and partner researchers and provider specialists. Specific steps in at least some exemplary data collection processes are described hereafter. The collected original data is stored in the lake database **170** as raw original data (e.g., documents, images, records, files, etc.).

[0202] At process block **406**, at least a subset of the collected data is "shaped" or otherwise processed to generate structured data that is optimal for database access, searching, processing and manipulation. Here, the data shaping process may take many forms and may include a plurality of data processing steps that ultimately result in optimal system structured data sets. At step **408** the database optimized shaped data is added to similarly structured data already maintained in data vault database **180**.

[0203] Continuing, at block **410**, at least a subset of the data vault data or the lake data is "shaped" or otherwise processed to generate structured data that is optimal to support specific user application programs **188** and **192** (see again FIG. **2**). Here, again, the data shaping process may take many forms and may include a plurality of data processing steps that ultimately result in optimal application supporting structured data sets. At step **412** the optimized application structured data is added to similarly structured data already maintained in data marts database **190**.

[0204] Referring again to FIG. **4**, at block **414**, system users employ various application programs to access and manipulate system data including the data in any of the lake database **170**, data vault database **180** and data marts **190**. At block **212**, as users use the system, data related to system use is collected after which control passes backup to block **206** where the collected use data is shaped and eventually stored for driving additional applications.

[0205] FIG. **5** includes a flow chart illustrating a process **500** that is consistent with at least some aspects of the present disclosure for ingesting initial raw data into the disclosed system. At process block **502** new raw data is received at the file gateway **314** (see FIG. **2**) which, at block **504**, determines whether or not the data should be rejected or ingested based on the data source, data format or other transport data used to transmit the received data to the gateway. If the data is to be ingested, gateway **312** gleans metadata from the received data at block **506** which is stored in the data lake catalog **226** (see FIG. **2**) while the received data set is stored in data lake **170** at **508**. At block **510**, an alert is added to the alert list **169** indicting the new data is available to be consumed along with a data type so that other data consumers can recognize when to consume the newly stored data. Control passes back up to block **502** where the process described above continues.

[0206] FIG. **6** is a flow chart illustrating a general process **600** by which system micro-services consume lake data and generate micro-service data products that are published back to the lake database for further consumption by other micro-services. At process block **602** a micro-service process is specified that includes data consumption and data product definitions as well as micro-service code for carrying out process steps. At block **604** the micro-service monitors the data lake **170** for alerts specifying new data that meets the data consumption definition for the specific micro-service. At block **606**, where new lake data alerts do not specify data that meets the data consumption definition, control passes back up to block **604** where steps **604** and **606** continue to cycle.

[0207] Referring still to FIG. **6**, once an alert indicates new data that meets the micro-service data consumption definition, control passes to block **608** where the micro-service accesses the lake data to be consumed and that data is consumed at block **610** which generates a new data product. Continuing, at block **612**, the new data product is published to data lake database **170** and at **614** another alert is added to the data alert list **169**.

[0208] Referring still to FIG. **6**, process **600** is associated with a single system micro-service. It should be understood that hundreds and in some cases even thousands of micro-services will be performed simultaneously and that two or more micro-services may be performed on the same raw data or using prior generated micro-service data product(s) at the same time. In many cases a micro-service will require two or more data sets at the same time and, in those cases, a micro-service will be programmed to monitor for all required data in the data lake and may only be initiated once all required data is indicated in the alerts list **169**.

[0209] As described above, some micro-services will be completely automated, so that no user activities are required, while other micro-services will require at least some user activities to perform some service steps. FIG. **7** illustrates a simple fully automated micro-service **700** while FIG. **8** illustrates a micro-service **800** where a user has to perform some activities. In FIG. **7**, at process block **702**, an OCR micro-service is specified that requires consumption of raw clinical medical records to generate semi-structured clinical medical records with OCR tags appended to document characters. At block **704** the OCR micro-service monitors the system alert list **169** for alerts indicating that new raw clinical records data is stored in the data lake.

[0210] At block **706**, where there is no new clinical record to be ingested into the system, control passes back up to block **704** and the process **700** cycles through blocks **704** and **706**. Once a new clinical record is saved to lake database **170** and an alert related thereto is detected by the OCR micro-service, the micro-service accesses the new raw clinical record from the data lake at **708** and that record is consumed at block **710** to generate a new OCR tagged record. The new OCR tagged record is published back to the lake at **712** and an alert related thereto is added to the data alert list **169** at **714**. Once the OCR tagged record is stored in lake database **170**, it can be consumed by other micro-services or other system modules or components as required.

[0211] The FIG. **8** process **800** is associated with a micro-service for generating a system optimized structured clinical record assuming that an unstructured clinical medical record that has already been tagged with medical terms, phrases and contextual meaning has been generated as a micro-service data product by a prior micro-service. At process block **802**, the record structuring micro-service process is defined and includes a data consumption definition that requires OCR, NLP records to be consumed and a data production definition where the system optimized data structure is generated as a micro-service data product. At block **804** the structuring micro-service listens for alerts that new records to consume have been stored in lake database **170**. At block **806**, where new data to consume has not been stored in the lake database **170**, control cycles back through blocks **804** and **806** continually. Once new data to consume has been stored in lake database **170**, control passes to block **808** where the micro-service places an alert in an abstractor specialist's work queue identifying the record to consume as requiring specialist activities to complete the micro-service.

[0212] Referring still to FIG. **8**, at block **810**, the system monitors for specialist selection of the queued record for consumption and the system cycles between blocks **808** and **810** until the record is selected. Once the record is selected by the abstractor specialist at **810**, control passes to block **812** where the record to be consumed is accessed in database **170**. At block **814**, the micro-service accesses a structured clinical record file which includes data fields to be populated with data from the accessed clinical record. The micro-service attempts to identify data in the clinical record to populate each field in the structured record at **814** and populates fields with data whenever possible to generate a structured clinical record draft.

[0213] Continuing, at block **816** a micro-service presents an abstractor application interface to the abstractor specialist that can be used to verify draft field entries, modify entries or to aid the abstractor specialist in identifying data to populate unfilled structured record fields. To this end, see

FIG. **9** that shows an exemplary abstractor interface screenshot **914** that may be viewed by an abstractor specialist which includes an original record in an original record field **900** on the right hand side of the shot and a structured record area **902** on the left hand side of the screenshot. The structured record in area **902** includes a set of fields to be populated with information from the original record or in some other fashion to prepare the structured record for use by system applications. The structured record shown in area **902** only shows a portion of the structured record that fits within area **902** and in most cases the structured record will have hundreds or even thousands of record fields that need to be populated with data. Exemplary structured record fields shown include a site field **904**, year fields **905** and a histology field **906**.

[0214] Referring still to FIG. **9**, the original record shown in field **900** has already been subjected to OCR and NLP so that words and phrases have been recognized by a system processor and the text in the document is associated with specific medical words and phrases or other meaning (e.g., dates are recognized as dates, a “Patient's Name” label on an original record is recognized as the phrase “patient's name” and an adjacent field is recognized as a field that likely includes a patient's name, etc.). Again, the processor examines the original record for data that can be used to populate the structured record fields in order to create at least a partially complete draft of the structured record for consideration and completion by the abstractor specialist.

[0215] Data in the original record used to populate any field in the structured record is highlighted (see **910**, **912**) or somehow visually distinguished within the original record to aid the abstractor specialist in located that data in the original record when reviewing data in the structured record fields. The specialist moves through the structured record reviewing data in each field, checking that data against the original record and confirming a match (e.g., via selection of a confirmation icon or the like) or modifying the structured record field data if the automatically populated data is inaccurate (see block **818** in FIG. **8**).

[0216] In cases where the processor cannot automatically identify data to populate one or more fields in the structured record, the specialist reviews the original record manually to attempt to locate the data required for the field and then enters data if appropriate data is located. Where the micro-service fills in fields that are then to be checked by the specialist, in at least some cases original record data used to populate a next structured record field to be considered by the specialist may be especially highlighted as a further aid to locating the data in the original record. In some cases the micro-service will be able to recognize data in several different formats to be used to fill in a structured record field and will be able to reformat that data to fill in the structured record field with a required form.

[0217] Referring again to FIG. **8**, at block **820**, once the structured clinical record has been completed, the complete system optimized structured clinical record is stored in lake database **170** and then a new data alert is added to alert list **169** at **822** to alert other micro-services and orchestration resources that the complete record is available to be consumed.

[0218] In some cases a system micro-service will “learn” from specialist decisions regarding data appropriate for populating different structured data sets. For instance, if a specialist routinely converts an abbreviation in clinical records to a specific medical phrase, in at least some cases the system will automatically learn a new rule related to that persistent conversion and may, in future structured draft records, automatically convert the abbreviation to its expanded form. Many other system learning techniques are contemplated.

[0219] In cases where a system micro-service can confirm structured record field information with high confidence, the micro-service may reduce the confirmation burden on the specialist by not highlighting the accurate information in the structured record. For instance, where a patient's date of birth is known, the micro-service may not highlight a patient DOB field in the structured record for confirmation.

[0220] Referring now to FIG. **10**, an exemplary multi-micro-service process **1000** for ingesting a clinical medical record and structuring the record optimally for database activities is illustrated. At step **1001**, a medical record is acquired in digital form. Here, where an original record is in paper form, acquiring a digital record may include scanning that record into the system via a scanner **1012** to generate a PDF or other digital representation which is then provided to a system server **150** for storage in database **160**. In other cases where the record is already in digital form (e.g., an EMR), the digital record can simply be stored by server **150** in database **160**.

[0221] A data normalization and shaping process is performed at **1002** that includes accessing an original clinical record from database **160** and presenting that record to a system specialist **40** as shown in FIG. **9**. As the original record is accessed or at some other prior time, an OCR micro-service **700** (see again FIG. **7**) is used to tag letters in the record. The tagged record is stored in the data lake and an alert is added to the alert list **169**. Next, an NLP micro-service **1008** accesses the OCR tagged record and performs an NLP process on the text in that record to generate an NLP processed record which is again stored in the data lake and another alert is added to the alert list **169**.

[0222] At **800** (see FIG. **8**), a draft structured clinical medical record is generated for the patient and is presented to an abstractor specialist via an interface as in FIG. **9** so that the specialist can correct errors.

[0223] Referring again to FIG. **10**, once the structured record has been filled in to the extent possible based on an original medical record, at block **1020** the specialist may perform some task to attempt to complete record fields that have not been filled. For instance, in a case where a specific structured record field cannot be filled based on information from the original record, the specialist may attempt to track down information related to the field from some other source. For example, in a simple case the specialist may call **1024** a physician that generated the original record to track down missing information. As another example, the specialist may access some other patient record (e.g., an insurance record, a pharmacy record, etc.) that may include additional information useable to populate an empty field. Once the structured record is as complete as possible, that record is stored at **1022** back to the system database **160**.

[0224] Referring now to FIG. **11**, an exemplary process **1100** for generating genomic patient and tumor data is illustrated. Robust nucleic acid extraction protocols and sequencing library construction protocols may be applied, and appropriately deep coverage across all targeted regions and appropriately designed analysis algorithms may be utilized. Prior to process **1100**, a genomic sequencing order may be received at file gateway **314** and, once ingested, may be stored in lake database **170** for subsequent consumption. Here, when a tumor sample corresponding to the sequencing order is received **1114**, the sample is associated with the order and process **1100** continues with the order being assigned to a lab technician's work queue to commence specimen sequencing **1116**. At **1116** the specimens are subjected to a genetic sequencing process using sequencing machine **1132** to generate genomic data for both the patient and the tumor specimens. At **1118** alterations from raw molecular data are called and at block **1120** pathogenicity of the variants is classified. At **1122** genomic phenotypes may be calculated. At **1123** an MSI assay may be performed. At **1124** at least a subset of the genomic data and/or an analysis of at least the subset of the genomic data is stored in system database **160**.

[0225] Referring still to FIG. **11**, different approaches may be utilized to implement the genetic sequencing process at **1116**. In one example, an oncology assay may be implemented that interrogates all or a subset of cancer-related genes in matched tumor and normal tissue. As used herein, “tumor” tissue or specimen refers to a tumor biopsy or other biospecimen from which the DNA and/or RNA of a cancer tumor may be determined. As used herein, “normal” tissue or specimen refers to a non-tumor biopsy or other biospecimen from which DNA and/or RNA may be determined. As used herein, “matched” refers to the tumor tissue and the normal tissue being correlated at the same position in a DNA and/or RNA sequence, such as a reference sequence. The assay may further provide whole transcriptome RNA sequencing for gene rearrangement detection. The assay may combine tumor and normal DNA sequencing panels with tumor RNA sequencing to detect somatic and germ line variants, as well as fusion mRNAs created from chromosomal rearrangements.

[0226] The assay may be capable of detecting somatic and germ line single nucleotide polymorphisms (SNPs), indels, copy number variants, and gene rearrangements causing chimeric mRNA transcript expression. The assay may identify actionable oncologic variants in a wide array of solid tumor types. The assay may make use of FFPE tumor samples and matched normal blood or saliva samples. The subtraction of variants detected in the normal sample from variants detected in the tumor sample in at least some embodiments provides greater somatic variant calling accuracy. Base substitutions, insertions and deletions (indels), focal gene amplifications and homozygous gene deletions of tumor and germline may be assayed through DNA hybrid capture sequencing. Gene rearrangement events may be assayed through RNA sequencing.

[0227] In one example, the assay interrogates one or more of the 1711 cancer-related genes listed in the tables shown in FIG. **22***a*-**22***j* (referred to herein as the “xE” assay). This targeted gene panel may be divided into a clinically actionable tier, wherein 130 tier **1** genes (see table in FIG. **23**) that can influence treatment decisions are assayed with an assigned detection cutoff of 5% variant allele fraction (VAF) i.e. the limit of detection is 5% VAF or lower, and a secondary tier, wherein an additional 1,581 genes (e.g., the difference between the gene set in FIGS. **22***a*-**22***j* and FIG. **23**) are assayed for analytical purposes with an assigned detection cutoff of 10% VAF (limit of detection 10% VAF or lower). The RNA based gene rearrangement detection may also be divided into a primary clinically-actionable tier containing 41 rearrangements (See table in FIG. **24**), and a secondary tier that may contain some or all known fusions within the wider literature or novel fusions of putative clinical importance detected by the assay. “Tier 1” genes are genes linked with response or resistance to targeted therapies, resistance to standard of care, or toxicities associated with treatment. The VAF cutoff percentages described herein are exemplary and other cutoff values may be utilized. Reads may be mapped to a human reference genome, such as hg16, hg17, hg18, hg19, etc. (available from the Genome Reference Consortium, at https://www.ncbi.nlm.nih.gov/grc). In another example, the assay may interrogate other gene panels, such as the panels listed in the tables shown in FIGS. **27***a*, **27***b***1**, **27***b***2**, **27***c***1** and **27***c***2** and **27***d* (herein “the xT panel”) or the panel listed in the table shown in FIGS. **28***a* and **28***b*.

[0228] Referring still to FIG. **11**, the alterations called in sub-process **1118** may be called through a clinical variant calling process. An exemplary variant calling process is shown in FIG. **11***a*. At **1134** acceptance criteria are applied to the raw molecular data for clinical variant calling. There may be one or more acceptance criteria, and multiple acceptance criteria may be applied.

[0229] One type of acceptance criteria is that a certain percentage of loci assay must exceed a certain coverage. For instance, a first percentage of loci must exceed a certain first coverage and a second percentage of loci must exceed a second coverage. The first percentage of loci may be 60%, 65%, 70%, 75%, 80%, 85%, etc. and the first coverage level may be 150×, 200×, 250×, 300×, etc. The second percentage of loci may be 60%, 65%, 70%, 75%, 80%, 85%, etc. and the second coverage level may be 150×, 200×, 250×, 300×, etc. The first percentage of loci assayed may be lower than the second percentage of loci assayed while the first coverage level may be deeper than the second coverage level.

[0230] Another type of acceptance criteria may be that the mean coverage in the tumor sample meets or exceeds a certain coverage threshold, such as 300×, 400×, 500×, 600×, 700×, etc.

[0231] Another type of acceptance criteria may be that the total number of reads exceeds a predefined first threshold for the tumor sample and a predefined second threshold for the normal sample. For instance, the total number of reads for the tumor sample must exceed 5 million, 10 million, 15 million, 20 million, 25 million, 30 million, 35 million, 40 million, etc. reads and the total number of reads for the normal sample must exceed 5 million, 10 million, 15 million, 20 million, 25 million, 30 million, 35 million, 40 million, etc. reads. In one example, the threshold for the total number of the reads for the tumor sample may be greater than the total number of reads for the normal sample. For instance, the threshold for the total number of the reads for the tumor sample may be greater than the total number of reads for the normal sample by 5 million, 10 million, 5 million, 10 million, 15 million, 20 million, 25 million, 30 million, 35 million, 40 million, etc. reads.

[0232] Another type of acceptance criteria is that reads must maintain an average quality score. The quality score may be an average PHRED quality score, which is a measure of the quality of the identification of the nucleobases generated by automated DNA sequencing. The quality score may be applied to a portion of the raw molecular data. For instance, the quality score may be applied to the forward read. Another type of acceptance criteria is that the percentage of reads that map to the human reference genome. For instance, at least 60%, 65%, 70%, 75%, 80%, 85%, 80%, 95%, etc. of reads must map to the human reference genome.

[0233] Still at **1134**, RNA acceptance criteria may additionally be reviewed. One type of RNA acceptance criteria is that a threshold level of read pairs will be generated by the sequencer and pass quality trimming in order to continue with fusion analysis. For instance, the threshold level may be 5 million, 10 million, 15 million, 20 million, 25 million, 30 million, 35 million, 40 million, etc. Another type of acceptance criteria is that reads must maintain an average quality score. The quality score may be an average RNA PHRED quality score, which is a measure of the quality of the identification of the nucleobases generated by automated RNA sequencing. The quality score may be applied to a portion of the raw molecular data. For instance, the quality score may be applied to the forward read.

[0234] Yet another type of acceptance criteria is that the percentage of reads that map to the human reference genome. For instance, at least 60%, 65%, 70%, 75%, 80%, 85%, 80%, 95%, etc. of reads must map to the human reference genome.

[0235] If RNA analysis fails pre or post-analytic quality control, DNA analysis may still be reported. Due to the difficulties of RNA-seq from FFPE, a higher than normal failure rate is expected. Because of this, it may be standard to report the DNA variant calling and copy number analysis section of the assay, no matter the outcome of RNA analysis.

[0236] At **1138**, the step of variant quality filtering may be performed. Variant quality filtering may be performed for somatic and germline variations. For somatic variant filtering, the variant may have at least a minimum number of reads supporting the variant allele in regions of average genomic complexity. For instance, the minimum number of reads may be 1, 2, 3, 4, 5, 6, 7, etc. A region of the genome may be determined free of variation at a percentage of LLOD (for instance, 5% of LLOD) if it is sequenced to at least a certain read depth. For instance, the read depth may be 100×, 150×, 200×, 250×, 300×, 350×, etc.

[0237] The somatic variant may have a minimum threshold for SNPs. For instance, it may have at least 20×, 25×, 30×, 35×, 40×, 45×, 50×, etc. coverage for SNPs. The somatic variant may have a minimum threshold for indels. For instance, at least 50×, 55×, 60×, 65×, 70×, 75×, 80×, 85×, 90×, 95×, 100×, etc. coverage for indels may be required. The variant allele may have at least a certain variant allele fraction for SNPs. For instance, it may have at least 1%, 3%, 5%, 7%, 9%, etc. variant allele fraction for SNPs. The variant allele may have at least a certain variant allele fraction for indels. For instance, it may have a 6%, 8%, 10%, 12%, 14%, etc. variant allele fraction for indels.

[0238] The variant allele may have at least a certain read depth coverage of the variant fraction in the tumor compared to the variant fraction in the normal sample. For instance, the variant allele may have 4×, 6×, 8×, 10× etc. the variant fraction in the tumor compared to the variant fraction in the normal sample. Another type of filtering criteria may be that the bases contributing to the variant must have mapping quality greater than a threshold value. For instance, the threshold value may be 20, 25, 30, 35, 40, 45, 50, etc.

[0239] Another type of filtering criteria may be that alignments contributing to the variant must have a base quality score greater than a threshold value. For instance, the threshold value may be 10, 15, 20, 25, 30, 35, etc. Variants around homopolymer and multimer regions known to generate artifacts may be filtered in various manners. For instance, strand specific filtering may occur in the direction of the read in order to minimize stranded artifacts. If variants do not exceed the stranded minimum deviation for a specific locus within known artifact generating regions, they may be filtered as artifacts.

[0240] Variants may be required to exceed a standard deviation multiple above the median base fraction observed in greater than a predetermined percentage of samples from a process matched germline group in order to ensure the variants are not caused by observed artifact generating processes. For instance, the standard deviation multiple may be 3×, 4×, 5×, 6×, 7×, etc. For instance, the predetermined percentage of samples may be 15%, 20%, 25%, 30%, 35%, etc.

[0241] Still at **1138**, for germline variant filtering, the germline variant may have a minimum threshold for SNPs. For instance, it may have at least 20×, 25×, 30×, 35×, 40×, 45×, 50×, etc. coverage for SNPs. The germline variant may have a minimum threshold for indels. For instance, at least 50×, 55×, 60×, 65×, 70×, 75×, 80×, 85×, 90×, 95×, 100×, etc. coverage for indels may be required. The germline variant calling may require at least a certain variant allele fraction. For instance, it may require at least 15%, 20%, 25%, 30%, 35%, 40%, 45% etc. variant allelic fraction.

[0242] Another type of filtering criteria may be that the bases contributing to the variant must have mapping quality greater than a threshold value. For instance, the threshold value may be 20, 25, 30, 35, 40, 45, 50, etc. Another type of filtering criteria may be that alignments contributing to the variant must have a base quality score greater than a threshold value. For instance, the threshold value may be 10, 15, 20, 25, 30, 35, etc.

[0243] At **1142**, copy number analysis may be performed. Copy number alteration may be reported if more than a certain number of copies are detected by the assay, such as 3, 4, 5, 6, 7, 8, 9, 10, etc. Copy number losses may be reported if the ratio of the segments is below a certain threshold. For instance, copy number losses may be reported if the log 2 ratio of the segment is less than −1.0.

[0244] At **1146**, RNA fusion calling analysis may be conducted. RNA fusions may be compared to information in a gene-drug knowledge database **1148**, such as a database described in “Prospective: Database of Genomic Biomarkers for Cancer Drugs and Clinical Targetability in Solid Tumors.” Cancer Discovery 5, no. 2 (February 2015): 118-23. doi: 10.1158/2159-8290.CD-14-1118. If the RNA fusion is not present within the gene-drug knowledge database **1148**, the RNA fusion may not be presented. RNA fusions may not be called if they display fewer than a threshold of breakpoint spanning reads, such as fewer than 2, 3, 4, 5, 6, 7, 8, 9, 10, etc. breakpoint spanning reads. If an RNA fusion breakpoint is not within the body of two genes (including promotor regions), the fusion may not be called.

[0245] At **1150**, DNA fusion calling analysis may be performed. At **1154**, joint tumor normal variant calling data may be prepared for further downstream processing and analysis. Germline and somatic variant data are loaded to the pipeline database for storage and reporting. For example, for both somatic and germline variations, the data may include information on chromosome, position, reference, alt, sample type, variant caller, variant type, coverage, base fraction, mutation effect, gene, mutation name, and filtering. FIG. **25** shows an exemplary data set in table form that is consistent with at least some embodiments of the above disclosure.

[0246] Copy Number Variant (CNV) data may also be loaded to the pipeline database for downstream analysis. For example, the data may include information on chromosome, start position, end position, gene, amplification, copy number, and log 2 ratios. FIG. **26** includes exemplary CNV data.

[0247] Following analysis, a workflow processing system may extract and upload the variant data to the bioinformatics database. In one example, the variant data from a normal sample may be compared to the variant data from a tumor sample. If the variant is found in the normal and in the tumor, then it may be determined that the variant is not a cause of the patient's cancer. As a result, the related information for that variant as a cancer-causing variant may not appear on a patient report. Similarly, that variant may not be included in the expert treatment system database **160** with respect to the particular patient. Variant data may include translation information, CNV region findings, single nucleotide variants, single nucleotide variant findings, indel variants, indel variant findings, variant gene findings. Files, such as BAM, FASTQ, and VCF files, may be stored in the expert treatment system database **160**.

[0248] Referring again to FIG. **11**, at **1123**, an MSI assay may be performed as a next generation sequencing based test for microsatellite instability. The MSI assay may comprise a panel of microsatellites that are frequently unstable in tumors with mismatch repair deficiencies to determine the frequency of DNA slippage events. Using the assay methods, tumors may be classified into different categories, such as microsatellite instability high (MSI-H), microsatellite stable (MSS), or microsatellite equivocal (MSE). The assay may require FFPE tumor samples with matched normal saliva or blood to determine the MSI status of a tumor. MSI status can provide doctors with clinical insight into therapeutic and clinical trial options for patient care, as well as the need for further genetic testing for conditions such as Lynch Syndrome. The MSI algorithm may be initiated after the raw sequencing data is processed through the bioinformatics pipeline. Upon completion of the MSI algorithm, results may be stored in the expert treatment system database **160**. U.S. Prov. Pat. App. No. 62/745,946, filed Oct. 15, 2018, incorporated by reference in its entirety, describes exemplary systems and methods for MSI algorithms.

[0249] Referring still to FIG. **11**, sub-processes **1116** through **1123** may be substantially or, in some cases even completely automated so that there is little if any lab technician activity required to complete those processes. In other cases each of the sub-processes **1116** through **1123** may include one or more lab technician activities and one or more automated micro-service steps or calculations. Again, in cases where a lab technician performs service steps, the micro-service may present instructions or other interface tools to help guide the technician through the manual service steps. At the end of each manual step some indication that the step has been completed is received by the micro-service. For instance, in some cases a system machine (e.g., the sequencing computer **1132**) may provide one or more data products to the micro-service that indicate completion of the step. As another instance, a technician may be queried for specific data related to the stage of the service. As yet one other instance, a technician may simply enter some status indication like, step completed, to indicate that process **1100** should continue.

[0250] One exemplary workflow **1153** with respect to the bioinformatics pipeline is shown in FIG. **11***b*. Referring also to FIG. **11***c*, a client, such as an entity that generates a bioinformatics pipeline, can register new samples **1157** and upload variant call text files **1159** for processing to a cloud service **1161**. The cloud service **1161** may initiate an alert by adding a message **1163** to a queue service **1165** (e.g., to an alert list) for each uploaded file. Input micro-services **1167** (**1167** in FIG. **11***c*) receive messages **1169** about each incoming file and process each of those files one at a time (see **1171**) as they are received to process and validate each file. The input micro-services **1167** may run as separate node processes and, in at least some cases, generate SQL insertion statements **1173** to add each validated file to the expert treatment system database **160**.

[0251] Referring still to FIGS. **11***b* and **11***c*, the input micro-services **1167** may also run a variant classification engine **1360** on the variant files utilizing a knowledge database of variant information **1175** to calculate many different types of variant criteria, further classification and addition database insertion. The variant micro-service **1167** may publish an alert **1183** when a key event occurs, to which other services **1179** can subscribe in order to react. After a variant call text file is parsed, the variant micro-service may insert variant analysis data into the expert treatment system database **160** including criteria, classifications, variants, findings, and sample information.

[0252] Other micro-services **1179** can query **1181** samples, findings, variants, classifications, etc. via an interface **1177** and SQL queries **1187**. Authorized users may also be permitted to register samples and post classifications via the other micro-services.

[0253] Referring to FIG. **12**, an organoid modelling process **1200** is illustrated that is consistent with at least some aspects of the present disclosure. At **1201** a tumor specimen **1230** is obtained which is divided into multiple specimens and each specimen is then grown **1202** as a 3D organoid **1232** in a special growth media designed to promote organoid development. At **1204** different cancer treatments are applied to each of the organoids to elicit responses. At **1206** a provider specialist observes the treatment results and at **1208** the results are characterized to assess efficacy of each treatment. At **1210** the results are stored in the system database **160** as part of the unified structured data set for the patient.

[0254] Referring to FIG. **13**, a process **1300** for ingesting radiological images into the disclosed system and for identifying treatment relevant tumor features is illustrated. At **1302** a set of 2D medical images including a tumor and surrounding tissue are either generated or acquired from some other source and are stored in system database **160** (e.g., as unaltered images in the lake database). In many cases the 2D images will be in a digital format suitable for processing by a system processor. In other cases the 2D images will be in a format that has to be converted to a data set suitable for system analysis. For instance, in some cases the original images may be on film and may need to be scanned into a digital format prior to creating a 3D tumor model. In some cases original images may not be useable to generate a 3D tumor model and in those cases additional imaging may be required to generate the model.

[0255] At **1304** tumor tissue is detected and segmented within each of the 2D images so that tumor tissue and different tissue types are clearly distinguished from surrounding tissues and substances and so that different tumor tissue types are distinguishable within each image. At **1306** the tissue segments within the 2D images are used as a guide for contouring the tissue segments to generate a 3D model of the tumor tissue. At **908** a system processor runs various algorithms to examine the 3D model and identify a set of radiomic (e.g., quantitative features based on data characterization algorithms that are unable to be appreciated via the naked eye) features of the segmented tumor tissue that are clinically and/or biologically meaningful and that can be used to diagnose tumors, assess cancer state, be used in treatment planning and/or for research activities. At **1310** the 3D model and identified features are stored in the system database **160**.

[0256] While not shown, in some cases a normalization process is performed on the medical images before the 3D model is generated, for example, to ensure a normalization of image intensity distribution, image color, and voxel size for the 3D model. In other cases the normalization process may be performed on a 3D model generated by the disclosed system. In at least some cases the system will support many different segmentation and normalization processes so that 3D models can be generated from many different types of original 2D medical images and from many different imaging modalities (e.g., X-ray, MRI, CT, etc.). U.S. provisional patent application No. 62/693,371 which is titled “3D Radiomic Platform For Managing Biomarker Development” and which was filed on Jul. 2, 2018 teaches a system for ingesting radiological images into the disclosed system and that reference is incorporated herein in its entirety by reference.

[0257] Referring again to FIG. **11***c*, a therapy matching engine **1358** may match therapies based on the information stored in database **160**. In one example, the therapy matching engine **1358** matches therapies at the gene level and uses variant-level information to rank the therapies within a case. For each variant in a case, the therapy matching engine **1358** retrieves therapies matching a variant gene from an actionability database **1350**. The actionability database **1350** may store a variety of information for different kinds of variants, such as somatic functional, somatic positional, germline functional, germline positional, along with therapies associated with SNVs and indels.

[0258] Therapy matching engine **1358** may rank therapies for each gene based on one or more factors. For instance, the therapy matching engine may rank the therapies based on whether the patient disease (such as pancreatic cancer) matches the disease type associated with the therapy evidence, whether the patient variant matches the evidence, and the evidence level for the therapy. For CNVs, the therapy matching engine may automatically determine that the patient variant matches the evidence. For SNVs or indels, the therapy matching engine may evaluate whether the therapy data came from a functional input or a positional input. For positional SNV/indels, if a variant value falls within the range of the variant locus start and variant locus end associated with the evidence, the therapy matching engine may determine that the patient variant matches the evidence. The variant locus start and variant locus end may reflect those locations of the variant in the protein product (an amino acid sequence position).

[0259] For functional SNV/indels, if a variant mechanism matches the mechanism associate with the evidence, the therapy matching engine may determine that the patient variant matches the evidence. Therapies may then be ranked by evidence level. The first level may be “consensus” evidence determined by the medical community, such as medical practice guidelines. The next level may be “clinical research” evidence, such as evidence from a clinical trial or other human subject research that a therapy is effective. The next level may be “case study” evidence, such as evidence from a case study published in a medical journal. The next level may be “preclinical” evidence, such as evidence from animal studies or in vitro studies. Ultimately, pdf or other format reports **1368** are generated for consumption.

[0260] While a set of data sources and types are described above, it should be appreciated that many other data sets that may be meaningful from a research or treatment planning perspective are contemplated and may be accommodated in the present system to further enhance research and treatment planning capabilities.

[0261] Referring now to FIG. **3***a*, a schematic is shown that represents an exemplary data platform **364** that is consistent with at least some aspects of the present disclosure. The exemplary platform shows data, information and samples as they exist throughout a system where different system processes and functions are controlled by different entities including an overall system provider that operates both single tenant and multi-tenant cloud service platforms **368** and **372**, respectively, partners **366** that provide clinical files as well as tissue samples and related test requisition orders as well as other partners **374** that access processed data and information stored on the service platforms **368** and **372**. Partners **366** provide secure clinical files **375** via a file transfer to the single tenant cloud platform **368** and are stored as unstructured and identified files in the lake database. Those files are abstracted and shaped as described above to generate normalized structured clinical data that is stored in a single tenant data vault as well as in a multi-tenant data vault **388**. The data from the vault is then de-identified and stored in a de-identified clinical data database which is accessible to authorized partners **374** via system interfaces **383** and applications **381** as described herein.

[0262] Referring still to FIG. **3***a*, partners **366** also provide tissue samples and test requisition orders that drive next generation sequencing lab activity at **385** to generate the bioinformatics pipeline **386** which is stored in both a molecular data lake database **389** and the multi-tenant data vault **388**. The data in vault **388** is de-identified and stored in an aggregate de-identified clinical data database **390** where it is accessible to authorized partners via system interfaces **393** and applications **382** as described herein. In addition, the molecular lake data **389** and the de-identified single tenant files **380** are accessible to other authorized partners via other interfaces **384**.

IV. User Interfaces

[0263] Referring again to FIG. **3**, the disclosed system **100** is accessible by many different types of system users that have many different needs and goals including clinical physicians **10** as well as provider specialists like data abstractors **20**, lab, modeling and radiology specialists **30**, partner researchers **40**, provider researchers **50** and dataset sales specialists **60**, among others. Because each user type performs different activities aimed at achieving different goals, the application suites **188**, **192** and associated user interfaces employed by each user type will typically be at least somewhat if not very different. For instance, a physician's application suite may include 9 separate application programs that are designed to optimally support many oncological treatment consideration and planning processes while an abstractor specialist's application suite may include 5 application programs that are completely separate from the 9 programs in the physician's suite and that are designed to optimally facilitate record abstraction and data structuring processes.

[0264] In some cases a system user's program suite will be internally facing meaning that the user is typically a provider employee and that the suite generates data or other information deliverables that are to be consumed within the system **100** itself. For instance, an abstractor application program for structuring data from a raw data set to be consumed by micro-services and other system resources is an example of an internally facing application program. Other system user programs or suites will be externally facing meaning that the user is typically a provider customer and that the suite generates data or other information deliverables that are primarily for use outside the system. For instance, a physician's application program suite that facilitates treatment planning is an example of an externally facing program suite.
[0265] Referring now to FIGS. **14** through **21**, screenshots of an exemplary physician's user interface that include a series of hyperlinked user interface views that are consistent with at least some aspects of the present disclosure are shown. The screenshots show one natural progression of information consideration wherein each interface is associated with one of the physician's program suite applications **188**. While some of the illustrated screenshots are complete, others are only partial and additional screen data would be accessible via either scrolling downward as well known in the graphical arts or by selection of a hyperlink within the presented view that accesses additional information related to the screenshot that includes the selected hyperlink.
[0266] Referring to FIG. **14**, once a physician logs onto system **10** via entry of a username and password or via some other security protocol, the physician is either presented with a patient list screen **1400** or can navigate to that screen. The patient list screen **1400** includes a first navigation bar or ribbon that extends along an upper edge of the view as well as a patient list area **1405** that includes a separate cell or field (two labelled **1402** and **1404**) for each of the physician's patients for which the system **100** stores data. Each patient cell (e.g., **1404**) includes basic patient information including the patient's name, an identification number and a cancer type and operates as a hyperlink phrase for accessing applications where the system loads data for the patient indicated in the cell. The screen **1400** also includes a “New Patient” icon **1406** that is selectable to add a new patient to the physician's view. The screen **1400** may display all patients of the physician's who have received genomic testing. Each patient cell can represent one or more reports created based on tissue samples. Physicians can also see in-progress patients along with a status indicating an order's progress, such as if the sample has been received. Some physicians may be provided with an additional section displaying reference patients. In these cases, the physician signed into the system **10** is not the patient's ordering physician, but has some other reason to access the patient information, such as because the ordering physician indicated he or she should receive a copy of the report and be permitted other appropriate access. Certain users of the system **10**, such as administrators, may have access to browse all patients within their institution.
[0267] Referring again to FIG. **14**, upon selecting cell **1404** associated with a patient named Dwayne Holder, the system presents the screenshot **1500** shown in FIG. **15** that includes a second level navigation bar **1502** near the top of the screen **1500** and a workspace **1504** below bar **1502**. Navigation bar **1502** persistently identifies the patient **1506** associated with the data currently being viewed by the physician throughout the screenshots illustrated and also includes a separate hyperlink text term for each of several system data views or application programs that can be selected by the physician. In FIG. **15** the view and applications options include an “Overview” option **1508**, a “Reports” option **1510**, an “Alterations” option **1512**, a “Trials” option **1514**, an “Immunotherapy” option **1516**, a “Cohort” option **1518**, a “Board” option **1520** and a “Modelling” option **1522**. Many other options will be added to bar **1502** over time as they are developed. A view or application currently accessed by the physician is underlined or otherwise visually distinguished in bar **1502**. For instance, in FIG. **15** the overview icon **1508** is shown highlighted to indicate that the information presented in workspace **1504** is associated with the overview data view.
[0268] Referring still to FIG. **15**, the exemplary overview view includes a patient care timeline **1509** along a left edge of workspace **1504**, high level patient cancer state information **1550** in a

central portion of workspace **1504** and view selection icons **1540** along a right edge of workspace **1504**. Timeline **1509** includes a set of patient care cells **1570**, **1580**, etc., each of which corresponds to a meaningful care related event associated with treatment of the patient's cancer state. The cells are vertically stacked with earliest cells in time near the bottom of the stack and more recent cells near the top of the stack. Each cell is typically restricted to activities or information associated with a specific date and, in addition to the associated date, may include any subset of several different information types including hospital or clinic admission and release dates, medical imaging descriptors, procedure descriptors, medication start and end dates, treatment procedure start and end descriptors, test descriptors, test or procedure results descriptors and other descriptors. This list is exemplary and not intended to be exhaustive. For instance, cell **1532** that is dated Dec. 29, 2017 indicates that a lung biopsy occurred as well as a brain CT imaging session and an MRI of the patient's abdomen. Information in the timeline **1509** may be loaded from the structured data that results from using the systems and methods described herein, such as those with reference to FIG. **10**. Information in the timeline **1509** may also include references to genomic sequencing tests ordered for a patient.

[0269] Referring still to FIG. **15**, in addition to including the patient care cell stack, the care timeline **1509** includes a vertical activity icon progression **1534** that extends along the left edge of the cell stack. The activity icons in progression **1534** are horizontally aligned with associated textual descriptions of care events in the cell stack. Each activity icon is designed to glanceably indicate an activity type so that a physician can quickly identify activities of specific types within the stacked cells by simply viewing the icons and associated stack event descriptors. For instance, exemplary activity icons include a gene panel publication icon **1552**, a medication start/stop icon **1554**, a facility admit/release icon **1556** and an imaging session icon **1558**. Other icons corresponding to surgery, detected patient medical conditions, and other procedures or important medical events are contemplated.

[0270] Referring still to FIG. **15**, in at least some cases detailed data related to a care event will be further accessible by selecting one of the activity icons along the left of the cells or events in a cell to hyperlink to the additional information. For instance, the “CT: Brain” text at **1662** may be selectable to link to a CT image viewer to view CT images of the patient's brain that correspond to the event. Other links are contemplated.

[0271] Referring again to FIG. **15**, general cancer state and patient information at **1550** includes diagnosis, stage, patient date of birth and gender information **1530** as well as an anatomical image that shows a representation of a tumor within a body that is generally consistent with the patient's cancer state. In some cases the tumor representation is just representative of the patient's condition as opposed to directly tied to actual tumor images while in other cases the tumor representation is derived from actual medical images of the patient's tumor.

[0272] Referring again to FIG. **15**, the patient body image **1550** may be overlaid with structured contours **1560** from the patient's radiology imaging. Represented structures may include primary or metastatic lesions, organs, edema, etc. A physician may click each structured contour to obtain an additional level of detail of information. Clicking the structured contour may isolate it visually for the physician. In the case of a tumor contour, the additional level of detail may include supporting information such as tumor volume, longest 3D diameter, or other features. Certain radiomic features that may be presented to the physician are described in further detail in, for instance, U.S. Provisional Patent Application No. 62/693,371, titled 3D Radiomic Platform for Imaging Biomarker Development, which has been incorporated herein by reference in its entirety.

[0273] From this detailed view, the physician may further drill down to an additional, microscopic level of detail. Here, a patient's histopathology results may be displayed. Clinical interpretations are shown, where available from an issued report. The microscopic detail may also display thumbnail images of microscope slides of a patient's specimens.

[0274] View selection icons **1540** include a set of icons that allow the physician to select different views of the patient's cancer condition and are progressively more granular. To this end, the exemplary view icons include a body view icon **1572** corresponding to the body view shown in FIG. **15**, a medical imaging view icon **1574** for accessing medical X-ray, CT, MRI and other images, a cellular view icon **1576** that shows cellular level images and genomic sequencing data icon **1578** for accessing genomic data views.

[0275] Referring again to FIG. **15**, to access specific issued reports associated with the patient the physician selects reports icon **1510** to access a reports screen **1600** shown in FIG. **16**. Reports screen **1600** shows the reports icon **1510** highlighted to help orient the physician and includes a report list indicating all reports stored in the system that are associated with the patient. In the exemplary reports view, each report is represented in the list by a reduced size image of the first page of the report and with a general report description field near the bottom of the image. For exemplary report images are shown at **1602** and **1604** and a general report description of the report associated with image **1602** is provided at **1606** indicating report type, date and other characterizing information.

[0276] The physician can select one of the report images to access the full report. For instance, if the physician selects image icon **1602**, the screenshot **1700** shown in FIG. **17** is presented that splits the display screen into a report list section **1702** along the left edge of the screen and an enlarged report section **1704** that covers about the right two thirds of the screen where the selected report is presented in a larger format for viewing. The report presents clinically significant information and may take many different forms. Each report is listed again in section **1702** as a reduced size hyper linkable image as shown at **1602** and **1604** where the currently selected report **1602** is highlighted or otherwise visually distinguished. The physician can select a PDF icon **1708** to download a copy of the report to the physician's computer.

[0277] A patient may have multiple reports for each specimen or specimen set sequenced. Reports may include DNA sequencing reports, IHC staining reports, RNA expression level reports, organoid growth reports, imaging and/or radiology reports, etc. Each report may contain results of sequencing of the patient's tumor tissue and, where available the normal tissue as well. Normal tissue can be used to identify which alterations, if any, are inherited versus those that the tumor uniquely acquired. Such differentiation often has therapeutic implications.

[0278] FIG. **17***a* shows an exemplary first page of a report screenshot indicating the results of one RNA sequencing process. Profiling of whole RNA transcriptome provides molecular information that is complementary to DNA sequencing and can be clinically important to physicians. For example, RNA sequencing can assist in clinically validated unbiased translocation detection. Overexpression and underexpression of certain genes may be presented to the physician as a result of RNA sequencing. Likewise, treatment implications may be provided to the physician which the physician may take into consideration when determining the best type of treatment for a patient. The physician may decide to verify results, for instance, through an orthogonal assay methodology, before using the results in clinical decision making.

[0279] To examine information related to a patient's genomic tumor alterations and possible treatment options, the physician selects alterations icon **1512** to access screen **1800** shown in FIG. **18**. Screen **1800** includes an approved therapies list **1802** and a pertinent genes list **1804**. The therapies list **1802** includes a list of genes for which variants have been identified and for each gene in the list, the associated variant, how the variant is indicated and other information including details regarding considerations corresponding to the associated therapy option. Other screens for considering alterations are contemplated to enable a physician to consider many aspects of treatment efficacy. Additional details may be provided to add context to alterations, such as gene descriptions, explanation of mutation effect, and variant allelic fraction. Alterations may be reported by category, ranging from highly relevant genes to variants of unknown significance.

[0280] Selecting an alteration may take the physician to an additional view, shown at FIGS. **18***a* and **18***b* (showing different scrolled sections of one view in the two figures), where the physician can delve deeper into the alteration's effect, with supporting data visualizations. Germline alterations associated with diseases may be reported as incidental findings. In FIG. **18***a*, approved therapies are listed with relevant related information including a gene and variant indicator along with hyperlinks to evidence associated with the therapy and details about each of the therapies.

[0281] The physician application suite also provides tools to help the physician identify and consider clinical trials that may be related to treatment options for his patient. To access the trials tools, the physician selects trials icon **1514** to access the screen (not shown) that lists all clinical trials that may be of any interest to the physician given patent cancer state characteristics. For instance, for a patient suffering from pancreatic cancer, the list may indicate 12 different trials occurring within the United States. In some cases the trials may be arranged according to likely most relevant given detailed cancer state factors for the specific patient. The physician can select one of the clinical trials from the list to access a screen **1900** like the one shown in FIG. **19**. Screen **1900** includes a map **1904** with markers (three labelled **1906**, **1908** and **1910**) at map locations corresponding to institutions are participating in the selected trial as well as a general description **1920** of the trial. Screen **1900** also provides a set of filtering tools **1930** in the form of pull down menus the physician can use to narrow down trial options by different factors including distance from the patient's location, trial phase (e.g., not yet initiated, progressing, wrapping up, etc.), and other factors. Here, the idea is that the physician can explore trial options for specific patient cancer states quickly by focusing consideration on the most relevant and convenient trial options for specific patients.

[0282] The physician application suite provides tools for the physician to consider different immunotherapies that are accessible by selecting immunotherapy icon **1516** from the navigation bar. When icon **1516** is selected, an exemplary immunotherapy screenshot **2000** shown in FIG. **20** is presented. Screenshot **2000** includes a menu of immunotherapy interface options **2002** extending vertically along a left area of the screen and a detailed information area **2004** to the right of the options **2002**. In at least some cases the immunotherapy options **2002** will include a summary option, a tumor mutation burden option, a microsatellite instability status option, an immune resistance risk option and an immune infiltration option where each option is selectable to access specific immunotherapy data related to the patient's case. Immunotherapy options **2002** may provide the physician with an indication that an immunotherapy, such as an FDA approved immunotherapy, may be appropriate to prescribe the patient. Examples may include dendritic cell therapies, CAR-T cell therapies, antibody therapies, cytokine therapies, combination immunotherapies, adoptive t-cell therapies, anti-CD47 therapies, anti-GD2 therapies, immune checkpoint inhibitors, oncolytic viruses, polysaccharides, or neoantigens, among others. Area **2004** shows summary information presented when the summary option is selected from the option list **2002**. When other list options are selected, related information is used to populate area **2004** with additional related information.

[0283] Referring to FIG. **21**, the cohort option **1518** can be selected to access an analytical tool that enables the physician to explore prior treatment responses of patients that have the same type of cancer as the patient that the physician is planning treatment for in light of similarities in molecular data between the patients. To this end, once genomic sequencing has been completed for each patient in a set of patients, molecular similarities can be identified between any patients and used as a distance plotting factor on a chart **2110**. In FIG. **21**, the screen **2100** includes a graph at **2110**, filter options at **2120**, some view options **2140**, graph information at **2150** and additional treatment efficacy bar graphs at **2160**.

[0284] Referring still to FIG. **21**, the illustrated graph presents a tumor associated with the patient for which planning is progressing at a center location as a star and other patient tumors of a similar type (e.g., pancreatic) at different radial distances from the central tumor where molecular similarity is based on distance from the central location so that tumors more similar to the central tumor are near the center and tumors other than the central tumor are located in proximity to one another based on their respective similarity. Angular displacements between the other tumors represented indicate dissimilarity or similarity between any two tumors where a greater angular distance between two tumors indicates greater dissimilarity. Except for the central tumor (e.g., indicated via the star), each of the other tumors is color coded to indicate treatment efficacy. For instance, a green dot may represent a tumor that completely responded to treatment, a yellow dot may indicate a tumor that responded minimally while a red dot indicates a tumor that did not respond. An efficacy legend at **2130** is provided that associates tumor colors with efficacies "e.g., "Complete Response", "Partial Response", etc.), the physician can select different options to show in the graph including response, adverse reaction, or both using icons at **2140**.

[0285] Referring still to FIG. **21**, an initial view **2110** may include all patient tumors that are of the same general type as the central tumor presented on the graph **2110**, regardless of other cancer state factors. In FIG. **21**, a number "n" is equal to 975 indicating that 975 tumors and associated patients are represented on graph **2110**. Filters at **2120** can be used by the physician to select different cancer state filter factors to reduce the n count to include patients that have other factors in common with the patient associated with the central tumor. For instance, patient sex or age or tumor mutations or any factor combination supported by the system may be used to filter n down to a smaller number where multiple factors are common among associated patients.

[0286] Referring again to FIG. **21**, the efficacy bar graphs **2160** present efficacy data for different treatment types. To this end, screen area **2160** presents a list of medications or combinations thereof that have been used in the past to treat the tumors represented in graph **2110**. A separate bar graph is provided for each of the treatment medications or combinations where each bar graph includes different length color coded sub-sections that show efficacy percentages. For instance, for Germcitabine, the bar graph **2170** may include a green section that extends 11% of the length of the total bar graph and a blue section that extends 5% of the length of the total bar graph to indicate that 11% of patients treated with Germcitabine experienced a complete response while 5% experienced only a partial response. Other color coded sections of bar **2170** would indicate other efficacies. The illustrated list only includes two treatment regimens but in most cases the list would be much longer and each list regimen would include its own efficacy bar graph.

IV. Automated Cancer State-Treatment-Efficacy Insights Across Patient Populations

[0287] Referring again to FIG. **21**, the cohort tool shown allows a physician to select different cancer state filters **2120** to be applied to the system database thereby changing the set of patients for which the system presents treatment efficacy data to help the physician explore effects of different factors on efficacy which is intended to lead to new treatment insights like factor-treatment-efficacy relationships. While powerful, this physician driven system is only as good as the physician that operates it and in many cases cancer state-treatment-efficacy relationships simply will not even be considered by a physician if clinically relevant state factors are not selected via the filter tools. While a physician could try every filter combination possible, time restraints would prohibit such an effort. In addition, while a large number of filter options could be added to the filter tools **2120** in FIG. **21**, it would be impractical to support all state factors as filter options so that some filter combinations simply could not be considered.

[0288] To further the pursuit of new cancer state-treatment-efficacy exploration and research, in at least some embodiments it is contemplated that system processors may be programmed to continually and automatically perform efficacy studies on data sets in an attempt to identify statistically meaningful state factor-treatment-efficacy insights. These insights can be confirmed by researchers or physicians and used thereafter to suggest treatments to physicians for specific cancer states.

V. Exemplary System Techniques And Results

[0289] The systems and methods described above may be used with a variety of sequencing panels. One exemplary panel, the 595 gene xT panel referred to above (See again the FIG. **27** series of figures), is focused on actionable mutations. Hereafter we present a description of various techniques and associated results that are consistent with aspects of the present disclosure in the context of an exemplary xT panel.

[0290] Techniques and results include the following. SNVs (single nucleotide variants), indels, and CNVs (copy number variants) were detected in all 595 genes. Genomic rearrangements were detected on a 21 gene subset by next generation DNA sequencing, with other genomic rearrangements detected by next generation RNA sequencing (RNA Seq). The panel also indicated MSI (microsatellite instability status) and TMB (tumor mutational burden). DNA tumor coverage was provided at 500× read sequencing depth. Full transcriptome was also provided by RNA sequencing, with unbiased gene rearrangement detection from fusion transcripts and expression changes, sequenced at 50 million reads.

[0291] In addition to reporting on somatic variants, when a normal sample is provided, the assay permits reporting of germline incidental findings on a limited set of variants within genes selected based on recommendations from the American College of Medical Genetics (ACMG) and published literature on inherited cancer syndromes.

Mutation Spectrum Analysis for Exemplary 500 Patient xT Group

[0292] Subsequent to selection, patients were binned by pre-specified cancer type and filtered for only those variants being classified as therapeutically relevant. The gene set was then filtered for only those genes having greater than 5 variants across the entire group so as to select for recurrently mutated genes. Having collated this set, patients were clustered by mutational similarity across SNPs, indels, amplifications, and homozygous deletions. Subsequently, mutation prevalence data for the MSKCC IMPACT data were extracted from the MSKCC Cbioportal in order to compare the xT assay variant calls against publicly available variant data for solid tumors. After selecting for only those genes on both panels, variants with a minimum of 2.5% prevalence within their respective group were plotted.

Detection of Gene Rearrangements from DNA by the xT Assay

[0293] Gene rearrangements were detected and analyzed via separate parallel workflows optimized for the detection of structural alterations developed in the JANE workflow language. Following de-multiplexing, tumor FASTQ files were aligned against the human reference genome using BWA (Li et al., 2009). Reads were sorted and duplicates were marked with SAMBIaster (Faust et al., 2014). Utilizing this process, discordant and split reads are further identified and separated. These data were then read into LUMPY (Layer et al., 2014) for structural variant detection. A VCF was generated and then parsed by a fusion VCF parser and the data was pushed to a Bioinformatics database. Structural alterations were then grouped by type, recurrence, and presence within the database and displayed through a quality control application. Known and previously known fusions were highlighted by the application and selected by a variant science team for loading into a patient report.

Detection of Gene Rearrangements from RNA by the xT Assay

[0294] Gene rearrangements in RNA were analyzed via a separate workflow that quantitated gene level expression as well as chimeric transcripts via non-canonical exon-exon junctions mapped via split or discordant read pairs. In brief, RNA-sequencing data was aligned to GRCh38 using STAR (Dobin et al., 2009) and expression quantitation per gene was computed via FeatureCounts (Liao et al., 2014). Subsequent to expression quantitation, reads were mapped across exon-exon boundaries to un-annotated splice junctions and evidence was computed for potential chimeric gene products. If sufficient evidence was present for the chimeric transcript, a rearrangement was called as detected.

Gene Expression Data Collection

[0295] RNA sequencing data was generated from FFPE tumor samples using an exome-capture based RNA seq protocol. Raw RNA seq reads were aligned using CRISP and gene expression was quantified via the RNA bioinformatics pipeline. One RNA bioinformatics pipeline is now described. Tissues with highest tumor content for each patient may be disrupted by 5 mm beads on a Tissuelyser II (Qiagen). Tumor genomic DNA and total RNA may be purified from the same sample using the AllPrep DNA/RNA/miRNA kit (Qiagen). Matched normal genomic DNA from blood, buccal swab or saliva may be isolated using the DNeasy Blood & Tissue Kit (Qiagen). RNA integrity may be measured on an Agilent 2100 Bioanalyzer using RNA Nano reagents (Agilent Technologies). RNA sequencing may be performed either by poly (A)+transcriptome or exome-capture transcriptome platform. Both poly (A)+ and capture transcriptome libraries may be prepared using 1˜2 μg of total RNA. Poly (A)+RNA may be isolated using Sera-Mag oligo (dT) beads (Thermo Scientific) and fragmented with the Ambion Fragmentation Reagents kit (Ambion, Austin, TX). cDNA synthesis, end-repair, A-base addition, and ligation of the Illumina index adapters may be performed according to Illumina's TruSeq RNA protocol (Illumina). Libraries may be size-selected on 3% agarose gel. Recovered fragments may be enriched by PCR using Phusion DNA polymerase (New England Biolabs) and purified using AMPure XP beads (Beckman Coulter). Capture transcriptomes may be prepared as above without the up-front mRNA selection and captured by Agilent SureSelect Human all exon v4 probes following the manufacturer's protocol. Library quality may be measured on an Agilent 2100 Bioanalyzer for product size and concentration. Paired-end libraries may be sequenced by the Illumina HiSeq 2000 or HiSeq 2500 (2×100 nucleotide read length), with sequence coverage to 40˜75M paired reads. Reads that passed the chastity filter of Illumina BaseCall software may be used for subsequent analysis. Further details of the pipeline raw read counts may be normalized to correct for GC content and gene length using full quantile normalization and adjusted for sequencing depth via the size factor method (see Robinson, D. R. et al. Integrative clinical genomics of metastatic cancer. Nature 548, 297-303 (2017)). Normalized gene expression data was log, base 10, transformed and used for all subsequent analyses.

Reference Database

[0296] Gene expression data generated (as previously described) was combined with publicly available gene expression data for cancer samples and normal tissue samples to create a Reference Database. For this analysis, we specifically include data from The Cancer Genome Atlas (TCGA) Project and Genotype-Tissue Expression (GTEx) project. Raw data from these publically available datasets were downloaded via the GDC or SRA and processed via an RNAseq pipeline (described above). In total 4,865 TCGA samples and 6,541 GTEx samples were processed and included as part of the larger Reference Database for this analysis. After processing, these datasets were corrected to account for batch effect differences between sequencing protocols across institutions (i.e. TCGA & and the Reference Database). For example, TCGA and GTEx both sequenced fresh, frozen tissue using a standard polyA capture based protocol.

Gene Expression Calling

[0297] For each patient, the expression of key genes was compared to the Reference Database to determine overexpression or underexpression. 42 genes for over- or under-expression based on the specific cancer type of the sample were evaluated. The list of genes evaluated can vary based on expression calls, cancer type, and time of sample collection. In order to make an expression call, the percentile of expression of the new patient was calculated relative to all cancer samples in the database, all normal samples in the database, matched cancer samples, and matched normal samples. For example, a breast cancer patient's tumor expression was compared to all cancer samples, all normal samples, all breast cancer samples, and all breast normal tissue samples within the Reference Database. Based on these percentiles criteria specific to each gene and cancer type to determine overexpression was identified.

T-Distributed Stochastic Neighbor Embedding (T-SNE) RNA Analysis

[0298] The t-SNE plot was generated using the Rtsne package in R [R version 3.4.4 and Rtsne version 0.13] based on principal components analysis of all samples (N=482) across all genes (N=17,869). A perplexity parameter of 30 and theta parameter of 0.3 was used for this analysis.

Cancer Type Prediction

[0299] A random forest model was used to generate cancer type predictions. The model was trained on 804 samples and 4,526 TCGA samples across cancer types from the Reference Database. For the purposes of this analysis, hematological malignancies were excluded. Both datasets were sampled equally during the construction of the model to account for differences in the size of the training data. The random forest model was calculated using the Ranger package in R [R version 3.4.4 and ranger_0.9.0]. Model accuracy was calculated within the training dataset using a leave-one-out approach. Based on this data, the overall classification accuracy was 81%.

Tumor Mutational Burden (TMB)

[0300] TMB was calculated by determining the dividend of the number of non-synonymous mutations divided by the megabase size of the panel (2.4 MB). All non-silent somatic coding mutations, including missense, indel, and stop loss variants, with coverage greater than 100×and an allelic fraction greater than 5% were included in the number of non-synonymous mutations.

Human Leukocyte Antigen (HLA) Class I Typing

[0301] HLA class I typing for each patient was performed using Optitype on DNA sequencing (Szolek 2014). Normal samples were used as the default reference for matched tumor-normal samples. Tumor sample-determined HLA type was used in cases where the normal sample did not meet internal HLA coverage thresholds or the sample was run as tumor-only.

Neoantigen Prediction

[0302] Neoantigen prediction was performed on all non-silent mutations identified by the xT pipeline. For each mutation, the binding affinities for all possible 8-11aa peptides containing that mutation were predicted using MHCflurry (Rubinsteyn 2016). For alleles where there was insufficient training data to generate an allele-specific MHCflurry model, binding affinities were predicted for the nearest neighbor HLA allele as assessed by amino acid homology. A mutation was determined to be antigenic if any resulting peptide was predicted to bind to any of the patient's HLA alleles using a 500 nM affinity threshold. RNA support was calculated for each variant using varlens. Predicted neoantigens were determined to have RNA support if at least one read supporting the variant allele could be detected in the RNA-seq data.

Microsatellite Instability (MSI) Status

[0303] The exemplary xT panel includes probes for 43 microsatellites that are frequently unstable in tumors with mismatch repair deficiencies. The MSI classification algorithm uses reads mapping to those regions to classify tumors into three categories: microsatellite instability-high (MSI-H), microsatellite stable (MSS), or microsatellite equivocal (MSE). This assay can be performed with paired tumor-normal samples or tumor-only samples.

[0304] MSI testing in paired mode begins with identifying accurately mapped reads to the microsatellite loci. To be a microsatellite locus mapping read, the read must be mapped to the microsatellite locus during the alignment step of the exemplary xT bioinformatics pipeline and also contain the 5 base pairs in both the front and rear flank of the microsatellite, with any number of expected repeating units in between. All the loci with sufficient coverage are tested for instability, as measured by changes in the distribution of the number of repeat units in the tumor reads compared to the normal reads using the Kolmogorov-Smirnov test. If $p<=0.05$, the locus is considered unstable. The proportion of unstable loci is fed into a logistic regression classifier trained on samples from the TCGA colorectal and endometrial groups that have clinically determined MSI statuses.

[0305] MSI testing in unpaired mode also begins with identifying accurately mapped reads to the microsatellite loci, using the same requirements as described above. The mean number of repeat units and the variance of the number of repeat units is calculated for each microsatellite locus. A vector containing the mean and variance data for each microsatellite locus is put into a support vector machine classification algorithm trained on samples from the TCGA colorectal and endometrial groups that have clinically determined MSI statuses.

[0306] Both algorithms return the probability of the patient being MSI-H, which is then translated into a MSI status of MSS, MSE, or MSI-H.

Cytolytic Index (CYT)

[0307] CYT was calculated as the geometric mean of the normalized RNA counts of granzyme A (GZMA) and perforin (PRF1) (Rooney, M. S., Shukla, S. A., Wu, C. J., Getz, G. & Hacohen, N. Molecular and Genetic Properties of Tumors Associated with Local Immune Cytolytic Activity. Cell 160, 48-61 (2015)).

Interferon Gamma Gene Signature Score

[0308] Twenty-eight interferon gamma (IFNG) pathway-related genes (Ayers M., J Clin Invest 2017) were used as the basis for an IFNG gene. Hierarchical clustering was performed based on Euclidean distance using the R package ComplexHeatmap (version 1.17.1) and the heatmap was annotated with PD-L1 positive IHC staining, TMB-high, or MSI-high status. IFNG score was calculated using the arithmetic mean of the 28 genes.

Knowledge Database (KDB)

[0309] In order to determine therapeutic actionability for sequenced patients, a KDB with structured data regarding drug/gene interactions and precision medicine assertions is maintained. The KDB of therapeutic and prognostic evidence is compiled from a combination of external sources (including but not exclusive to NCCN, CIVIC {28138153}, and DGIdb {28356508}) and from constant annotation by provider experts. Clinical actionability entries in the KDB are structured by both the disease in which the evidence applies, and by the level of evidence. Therapeutic actionability entries are binned into Tiers of somatic evidence by patient disease matches as laid out by the ASCO/AMP/CAP working group {27993330}. Briefly, Tier I Level A (IA) evidence are biomarkers that follow consensus guidelines and match disease type. Tier I Level B (IB) evidence are biomarkers that follow clinical research and match disease type. Tier II Level C (IIC) evidence biomarkers follow the off-label use of consensus guidelines and Tier II Level D (IID) evidence biomarkers follow clinical research or case reports. Tier III evidence are variants with no therapies. Patients are then matched to actionability entries by gene, specific variant, patient disease, and level of evidence.

Alteration Classification, Prioritization, And Reporting

[0310] Somatic alterations are interpreted based on a collection of internally weighted criteria that are composed of knowledge of known evolutionary models, functional data, clinical data, hotspot regions within genes, internal and external somatic databases, primary literature, and other features of somatic drivers {24768039} {29218886}. The criteria are features of a derived heuristic algorithm that buckets them into one of four categories (Pathogenic/VUS/Benign/Reportable). Pathogenic variants are typically defined as driver events or tumor prognostic signals. Benign variants are defined as those alterations that have evidence indicating a neutral state in the population and are removed from reporting. VUS variants are variants of unknown significance and are seen as passenger events. Reportable variants are those that could be seen as diagnostic, offer therapeutic guidance or are associated with disease but are not key driver events. Gene amplifications, deletions and translocations were reported based on the features of known gene fusions, relevant breakpoints, biological relevance and therapeutic actionability.

[0311] For the tumor-only analysis germline variants were computationally identified and removed using by an internal algorithm that takes copy number, tumor purity, and sequencing depth into account. There was further filtering on observed frequency in a population database (positions with AF>1% ExAC non-TCGA group). The algorithm was purposely tuned to be conservative when calling germline variants in therapeutic genes minimizing removal of true somatic pathogenic alterations that occur within the general population. Alterations observed in an internal pool of 50 unmatched normal samples were also removed. The remaining variants were analyzed as somatic at a VAF>=5% and Coverage>=90. Using normal tissue, true germline variants were able to be flagged and somatic analysis contamination was evaluated. The Tumor/Normal variants were also set at the Tumor-only VAF/Coverage thresholds for analysis.

[0312] Clinical trial matching occurs through a process of associating a patient's actionable variants and clinical data to a curated database of clinical trials. Clinical trials are verified as open and recruiting patients before report generation.

Germline Pathogenic And Variants of Unknown Significance (VUS)

[0313] Alterations identified in the Tumor/Normal match samples are reported as secondary findings for consenting patients. These are a subset of genes recommended by the ACMG (Richards, S. et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet. Med. 17, 405-24 (2015)) and genes associated with cancer predisposition or drug resistance.

[0314] In an example patient group analysis, a group of 500 cancer patients was selected where each patient had undergone clinical tumor and germline matched sequencing using the panel of genes at FIGS. **27***a*, **27***b*, **27***c***1**, **27***c***2**, and **27***d* (known herein as the “xT” assay). In order to be eligible for inclusion in the group, each case was required to have complete data elements for tumor-normal matched DNA sequencing, RNA sequencing, clinical data, and therapeutic data. Subsequent to filtering for eligibility, a set of patients was randomly sampled via a pseudo-random number generator. Patients were divided among seven broad cancer categories including tumors from brain (50 patients), breast (50 patients), colorectal (51 patients), lung (49 patients), ovarian and endometrial (99 patients), pancreas (50 patients), and prostate (52 patients). Additionally, 48 tumors from a combined set of rare malignancies and 51 tumors of unknown origin were included for analyses for a total of nine broad cancer categories. These patients were collated together as a single group and used for subsequent group analyses.

[0315] The mutational spectra for the studied group was compared with broad patterns of genomic alterations observed in large-scale studies across major cancer types. First, data from all 500 patients was plotted by gene, mutation type, and cancer type, and then clustered by mutational similarity (FIG. **29**). The most commonly mutated genes included well-known driver mutations, including mutations in more than 5% of all cases in the group for TP53, KRAS, PIK3CA, CDKN2A, PTEN, ARIDIA, APC, ERBB2, EGFR, IDHI, and CDKN2B. These genes are known hallmarks of cancer and commonly found in solid tumors. Of these genes, CDKN2A, CDKN2B, and PTEN were most commonly found to be homozygously deleted, indicating loss-of-function mutations likely coinciding with loss of heterozygosity. These data demonstrate expected molecular signatures commonly seen in clinical solid tumor samples.

[0316] Previous pan-cancer mutation analyses have established mutational spectra within and across tumor types, and provide context to which the study group sequencing data may be compared. In FIG. **30**, the study group results were compared to a previously published pan-cancer analysis using the Memorial Sloan Kettering Cancer Center (MSKCC) IMPACT panel (Zehir, A. et al. Mutational landscape of metastatic cancer revealed from prospective clinical sequencing of 10,000 patients. Nat. Med. 23, 703-713 (2017)). In both datasets, we observed the same commonly mutated genes, including TP53, KRAS, APC and PIK3CA. These genes were observed at similar relative frequencies compared to the MSKCC group. These results indicate the mutation spectra within the study group is representative of the broader population of tumors that have been sequenced in large-scale studies.

[0317] Because both tumor and germline samples were sequenced in the group, the effect of germline sequencing on the accuracy of somatic mutation identification could be examined. Fiftyone cases were randomly selected from the study group with a range of tumor mutational burden profiles. Their variants were re-evaluated using a tumor-only analytical pipeline. After filtering the dataset using a population database and focusing on coding variants from the 51 samples, 2,544 variants were identified that had a false positive rate of 12.5%. By further filtering with an internally developed list of technical artifacts (e.g., artifacts from DNA sequencing process), an internal pool of matched normal samples, and classification criteria, 74% of the false somatic variants (false positive rate of 2.3%) were removed while still retaining all true somatic alterations.

[0318] To further characterize the tumors in the study group, RNA expression profiles for patients in the group were examined. Similar tumor types tend to have similar expression profiles (FIG. **31**). On average, samples within a cancer type as determined by pathologic diagnosis showed a higher pairwise correlation within the corresponding TCGA cancer group compared to between TCGA cancer groups (p-values=1Jun. 10, 2016). This clustering of samples by TCGA cancer group is observed in the t-SNE plot shown in FIG. **32**. For some tumor types, such as prostate cancer, metastatic samples cluster very closely to non-metastatic tumor samples. However other cancer types, most notably pancreatic cancer and colorectal cancer, form a distinct metastatic tumor cluster that also contains breast tumors and tumors of unknown origin. This effect is likely due to the effect of the background tissue on the expression profile of the tumor sample. For example, metastatic samples from the liver frequently, but not always, cluster together. This effect can also depend on the level of tumor purity within the sample.

[0319] Given the high-dimensionality of the data, we sought to determine whether we could predict cancer types using gene expression data. We developed a random forest cancer type predictor using a combination of publically available TCGA expression data and expression data generated at Tempus Labs. TCGA cancer type predictions compared to the xT group samples are shown in FIG. **32**. For example, 100% of breast cancer samples were correctly classified. Interestingly, using this method we are able to accurately classify these tumors even when the samples are biopsied from metastatic sites.

[0320] Additionally, it is notable that some of the “misclassified” samples may actually represent biologically and pathologically relevant classifications. For example, of the 50 brain tumors in our dataset, 48 (96%) were classified as gliomas, while 2 were classified as sarcomas.

[0321] One of these tumors carries a histopathologic diagnosis of “solitary fibrous tumor, hemangiopericytoma type, WHO grade III”, which is indeed a sarcoma. The other was diagnosed as “glioblastoma, WHO grade IV (gliosarcoma), with smooth muscle and epithelial differentiation”. The immunohistochemical profile is GFAP negative with desmin and SMA focally positive, supporting the diagnosis of gliosarcoma. It can be argued that the algorithm classified this tumor correctly by grouping it with sarcomas, and in fact, gliosarcomas carry a worse prognosis and have the ability to metastasize, differentiating them clinically from traditional glioblastoma.

[0322] Similarly, a case with a histopathologic diagnosis favoring carcinosarcoma was identified by the model as SARC in a patient with a history of prostate cancer presenting with a pelvic mass five years after surgery. The immunohistochemical profile of the tumor showed it was negative for the prostate markers prostatic acid phosphatase (PSAP) and prostatic specific antigen (PSA) and positive for SMA, consistent with sarcoma, which was thought to be secondary to prostate fossa radiation treatment. However, gene rearrangement analysis identified a TMPRSS2-ERG, suggesting that the tumor was in fact recurrent prostate cancer with sarcomatoid features.

[0323] The constellation of gene rearrangements and fusions in the study group were also examined. These types of genomic alterations can result in proteins that drive malignancies, such as EML4-ALK, which results in constitutive activation of ALK through removal of the transmembrane domain.

[0324] In order to assess assay decision support for clinically relevant genomic rearrangements, alterations detected using DNA or RNA sequencing assays were compared across assay type and for evidence matching them to therapeutic interventions. Overall, 28 total genomic rearrangements resulting in chimeric protein products were detected in the study group. 22 rearrangements were concordantly detected between assay type, four were detected via DNA-only assay, and two were detected via RNA-only assay (FIG. **33**). Of the three rearrangements detected via RNA sequencing, two of the three were not targets on the DNA sequencing assay and thus not expected to be detected via DNA sequencing. The functionality of these fusions were further analyzed via their predicted structures (FIGS. **34** and **35**). In all cases, algorithms predicted fully intact tyrosine kinase domains for RET and NTRK3 exemplar rearrangements, which may be potential therapeutic targets for tyrosine kinase inhibitors. This analysis indicates the utility of genomic rearrangement analyses as a source of clinically relevant information for therapeutic interventions.

[0325] To characterize the mutational landscape in all patients, the distribution of the mutational load across cancer types was analyzed. The median TMB across the study group was 2.09 mutations per megabase (Mb) of DNA with a range of 0-54.2 mutations/Mb.

[0326] The distribution of TMB varied by cancer type. For example, cancers that are associated with higher levels of mutagenesis, like lung cancer, had a higher median TMB (FIG. **36**). We found that there is a population of hypermutated tumors with significantly higher TMB than the overall distribution of TMB for solid tumors. These hypermutators are found in all cancer types, including cancers typically associated with low TMB, like glioblastoma (FIG. **36**). These hypermutated tumors are referred to as TMB-high, which are defined as tumors with a TMB greater than 9 mutations/Mb. This threshold was established by testing for the enrichment of tumors with orthogonally defined hypermutation (MSI-H) in a larger clinical database using the hypergeometric test. In this group, all MSI-H samples are in the TMB-high population (FIGS. **37** and **38**). The high mutational burdens from the remaining TMB-high samples were primarily explained by mutational signatures associated with smoking, UV exposure, and APOBEC mediated mutagenesis.

[0327] While TMB is a measure of the number of mutations in a tumor, the neoantigen load is a more qualitative estimate of the number of somatic mutations that are actually presented to the immune system. We calculated neoantigen load as the number of mutations that have a predicted binding affinity of 500 nM or less to any of a patient's HLA class I alleles as well as at least one read supporting the variant allele in RNA sequencing data. TMB was found to be highly correlated with neoantigen load ($R=0.933$, $p=2.42\times10^{-211}$) (FIG. **37**). This suggests that a higher tumor mutational burden likely results in a greater number of potential neoantigens.

[0328] The association of high TMB and MSI-H status with response to immunotherapy has been attributed to the greater immunogenicity of these highly mutated tumors. We used whole transcriptome sequencing to measure whether greater immunogenicity results in higher levels of immune infiltration and activation.

[0329] To test this, we assessed the relative levels of cytotoxic immune activity using a gene expression score, cytolytic index (CYT) (Rooney et al., 2015, “Molecular and Genetic Properties of Tumors Associated with Local Immune Cytolytic Activity,” Cell 160, pp. 48-61). We found that this two gene expression score is significantly higher in our TMB-high and MSI-high populations ($p=4.3\times10^{-5}$ and $p=0.015$, respectively) (FIG. **39**). This result demonstrates that even in patients with heavily pre-treated and advanced stage disease, a hypermutator status is strongly associated with greater cytotoxic immune activity.

[0330] Next, whether specific immune cell populations were differentially represented in the immune cell composition of TMB-high tumors compared to TMB-low was analyzed. We implemented a support vector regression-based deconvolution model to computationally estimate the relative proportion of 22 immune cell types in each tumor (Newman et al., 2015, “Robust enumeration of cell subsets from tissue expression profiles,” Nat. Methods 12, pp. 453-457. In accordance with our cytolytic index analysis, we also found that inflammatory immune cells, like CD8 T cells and M1 polarized macrophages, were significantly higher in TMB-high samples, while non-inflammatory immune cells, like monocytes, were significantly lower in TMB-low samples ($p=0.0001$, $p=2.8\times10^{-7}$, $p=0.0008$) (see FIG. **40**).

[0331] Increased immune pressure, like infiltration of more inflammatory immune cells, can lead tumors to express higher levels of immune checkpoint molecules like PD-L1 (CD274). These immune checkpoints function as a brake on the immune system, turning activated immune cells into quiescent ones. Accordingly, whole transcriptome analysis determined CD274 expression is significantly higher in the more immune-infiltrated TMB-high tumors ($p=0.0002$) (FIG. **41**).

CD274 expression is also highly correlated with the expression of its binding partner on immune cells, PDCD1 (PD-1), as well as other T cell lineage-specific markers like CD3E (FIG. **42**). Furthermore, samples that stained positive for PD-L1 protein via clinically-validated IHC tests cluster with higher CD274 RNA expression levels (FIG. **42**), suggesting the expression of CD274 may be used as a proxy for protein levels of PD-L1.

[0332] Transcriptomic markers were utilized to further determine whether patients that lack classically defined immunotherapy biomarkers still exhibited immunologically similar tumors. Using a 28 gene interferon gamma-related signature, it was found that tumor samples could be broadly categorized as either immunologically active “hot” tumors or immunologically silent “cold” tumors based on gene expression (FIG. **43**). The 28-gene set encompassed genes related to cytolytic activity (e.g., granzyme A/B/K, PRF1), cytokines/chemokines for initiation of inflammation (CXCR6, CXCL9, CCL5, and CCR5), T cell markers (CD3D, CD3E, CD2, IL2RG [encoding IL-2Ry]), NK cell activity (NKG7, HLA-E), antigen presentation (CIITA, HLA-DRA), and additional immunomodulatory factors (LAG3, IDO1, SLAMF6). Results support this stratification, with the immunologically “hot” population enriched for samples that were TMB-high, MSI-high or PDL1 IHC positive. Furthermore, TMB-high, MSI-high, or PD-L1 IHC positive tumors expressed higher levels of interferon gamma-related genes versus tumors without any of those biomarkers ($p=2.2\times10^{-5}$) (FIG. **44**). Hence, patients within this immunologically active cluster that lack traditional immunotherapy biomarkers represent an interesting patient population that may potentially benefit from immunotherapy.

[0333] The ultimate goal of the broad molecular profiling done in the xT assay is to match patients to therapies as effectively as possible, with targeted or immunotherapy options being the most desirable. We evaluated whether patients in the xT group matched to response and resistance therapeutic evidence based on consensus clinical guidelines by cancer type (see KDB in Methods). Across all cancer types, 90.6% matched to therapeutic evidence based on response to therapy (FIG. **56**), and 22.6% matched to evidence based on resistance to therapy (FIG. **57**).

[0334] For both response and resistance therapeutic evidence, approximately 24% of the group could be matched to a precision medicine option with at least a tier IB level. In particular, tier IA therapeutic evidence, as defined by joint AMP, ASCO, and CAP guidelines, was returned for 15.8% of patients (FIG. **58**). The maximum tier of therapeutic evidence per patient varied significantly by cancer type (FIG. **45**). For example, 58.0% of colorectal patients could be matched to tier IA evidence, the majority of which were for resistance to therapy based on detected KRAS mutations; while no pancreatic cancer patients could be matched to tier IA evidence. This is expected, as there are several molecularly based consensus guidelines in colorectal cancer, but fewer or none for other cancer types. Additionally, specific therapeutic evidence matches were made based on copy number variants (CNVs) (FIG. **46**) and single nucleotide variants (SNVs) and indels (FIG. **47**) for each cancer category.

[0335] Therapies were also matched to single gene alterations, either SNVs and indels or CNVs, and plotted by cancer type (FIG. **48**). Unfortunately, the two most commonly mutated genes in cancer are TP53 and KRAS, with TP53 only having Tier IIC evidence and drugs in clinical trials, and KRAS having Tier 1A evidence, but as resistance to therapies targeting other proteins (36 patients). However, many less commonly mutated genes have Tier 1A evidence for targeted therapies across a variety of cancer types. Notable in this category are the PARP inhibitors for BRCA1 and BRCA2 mutated breast and ovarian cancer (16 patients), which are currently also in clinical trials or being used off-label in other disease types harboring BRCA mutations, such as prostate and pancreatic cancer. The majority of the remaining targetable mutations with Tier 1A evidence are from the druggable portions of the MAP kinase cascade (MAPK/ERK pathway), including EGFR, BRAF and NRAS across colorectal and lung cancer (18 patients).

[0336] Therapeutic options were further matched based on RNA sequencing data. We focused on the expression of 42 clinically relevant genes selected based on their relevance to disease diagnosis, prognosis, and/or possible therapeutic intervention. Over or underexpression of these genes may be reported to physicians.

[0337] Expression calls were made by comparison of the patient tumor expression to the tumor and normal tissue expression in the data vault database **180** based on overall comparisons as well as tissue-specific comparisons. For example, each breast cancer case was compared to all cancer samples, all normal samples, all breast cancer samples, and all normal breast samples. At least one gene in 76% of patients with gene expression data was reported. The distribution of expression calls is shown by sample (FIG. **54**) and by gene (FIG. **55**). It was found that metastatic cases are equally as likely to have at least one reportable expression call compared to non-metastatic tumors (79% vs 75%, p-value=0.288). The most commonly reported gene is overexpression of MYC, which was seen in 80 (17%) patient tumors across the group. Next, the percent of patients with gene expression calls was determined and evidence for the association between gene expression and drug response (FIG. **49**) was identified. Among the cases with reported expression calls, 25% of cases across cancer types included evidence based on clinical studies, case studies, and preclinical studies reported in the literature.

[0338] Fusion proteins are proteins made from RNA that has been generated by a DNA chromosomal rearrangement, also known as a “fusion event.” Fusion proteins can be oncogenic drivers that are among the most druggable targets in cancer. Of the 28 chromosomal rearrangements detected in the study group, 26 were associated with evidence of response to various therapeutic options based on evidence tiers and cancer type (FIG. **50**). The majority of fusion events were TMPRSS-ERG fusions within prostate cancer patients in the group. TMPRSS-ERG fusions in prostate cancer were given a IID evidence level due to the early evidence around therapeutic response. Of the seven non-prostate cancer fusions, one was rated as evidence level IA, one was rated as IIC and five were rated evidence level IID. These detected fusions are clear drivers of cancer, part of consensus therapeutic guidelines and shown to be present with high sensitivity by the xT assay referred to herein.

[0339] Based on the immunotherapy biomarkers identified by the xT assays, we investigated what percentage of the group would be eligible for immunotherapy. We discovered 10.1% of the xT group would be considered potential candidates for immunotherapy based on TMB, MSI status, and PD-L1 IHC results alone (FIG. **51**). The number of MSI-high and TMB-high cases were distributed among cancer types. This represents the most common immunotherapy biomarkers measured in the group with 4% of patients positive for both TMB-high and MSI-high status. PD-L1 positive IHC alone were measured in 3% of the eligibility group, and was found to be the highest among lung cancer patients. TMB-high status alone was measured in 2.6% of the eligibility group, primarily in lung and breast cancer cases. PD-L1 positive IHC and TMB-high status was the minority of cases and measured in only 0.4% of the eligibility group.

[0340] Overall, clinically relevant molecular insights were uncovered for over 90% of the group based on SNVs, indels, CNVs, gene expression calls, and immunotherapy biomarker assays (FIG. **52**). The majority of therapeutic matches to patients were based on clinically relevant xT findings reported on SNVs and indels. This was followed by matches based on CNVs, gene expression calls, fusion detection, and immunotherapy biomarkers. In addition to therapeutic matching, we determined clinical-trial matching for the group based on molecular insights from the xT assay.

[0341] In total, 1952 clinical trials were reported for the xT 500 patient group. The majority of patients, 91.6%, were matched to at least one clinical trial, with 73.6% matched with at least one biomarker-based clinical trial for a gene variant on their final report. The frequency of biomarker-based clinical trial matches varied by diagnosis and outnumbered disease-based clinical trial matches (FIG. **53**). For example, gynecological and pancreatic cancers were typically matched to a biomarker-based clinical trial; while rare cancers had the least number of biomarker-based clinical trial matches and an almost equal ratio of biomarker-based to disease-based trial matching. The differences between biomarker versus disease-based trial matching appears to be due to the

frequency of targetable alterations and heterogeneity of those cancer types.
[0342] The particular embodiments disclosed above are illustrative only, as the invention may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope and spirit of the invention. Accordingly, the protection sought herein is as set forth in the claims below.
[0343] Thus, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the following appended claims.
[0344] To apprise the public of the scope of this invention, the following claims are made:

## Claims

**1**. A computer system for evaluating a cancer treatment efficacy for a human subject that is undergoing, or has completed, a cancer treatment for a cancer, the computer system comprising one or more processors, and a memory, wherein the memory stores instructions for performing a method using the one or more processors, the method comprising: A) obtaining, from an electronic data store, first genomic sequencing data from cell free DNA drawn from a blood sample of the human subject, wherein the first genomic sequencing data comprises a plurality of epigenetic patterns of cell free DNA within the blood sample; B) obtaining, from an electronic data store, second genomic sequencing data from cell free DNA drawn from the blood sample, wherein the second genomic sequencing data comprises sequence data of cell free DNA within the blood sample; C) mapping the first genomic sequencing data to corresponding locations in a reference human genome; D) mapping the second genomic sequencing data to corresponding locations in the reference human genome; E) determining an absence or presence of at least a first genomic alteration, from among a plurality of predefined genomic alterations, in the blood sample based on at least (i) the second genomic sequencing data, and (ii) the locations in the reference human genome to which the second genomic sequencing data has been mapped; and F) securely communicating, through a network connection, a structured data report that provides an assessment of the cancer treatment efficacy for the human subject responsive to at least the plurality of epigenetic patterns of cell free DNA within the blood sample and the absence or presence of at least the first genomic alteration in the blood sample.
**2**. The computer system of claim 1, wherein the method further comprising interfacing with a cloud service platform to perform at least one of the A) obtaining, B) obtaining, C) mapping, D) mapping, E) determining or F) securely communicating.
**3**. The computer system of claim 1, wherein the plurality of predefined genomic alterations consists of mutations in the genes listed in FIG. **22**.
**4**. The computer system of claim 1, wherein the plurality of predefined genomic alterations consists of mutations in the genes listed in FIG. **27**.
**5**. The computer system of claim 1, wherein the plurality of predefined genomic alterations consists of mutations in the genes listed in FIG. **28**.
**6**. The computer system of claim 1, wherein the first genomic alteration is a single nucleotide polymorphism in a gene.
**7**. The computer system of claim 1, wherein the first genomic alteration is an insertion or deletion mutation in a gene.
**8**. The computer system of claim 1, wherein the first genomic alteration is a genomic rearrangement occurring in a gene.
**9**. The computer system of claim 1, the method further comprising verifying that each genomic alteration in the plurality of predefined genomic alterations is a somatic genomic alteration.

**10**. The computer system of claim 1, wherein the assessment of the cancer treatment is in the form of a molecular residual disease status or a cancer recurrence status for the human subject, and wherein the method further comprises determining, for each respective genomic alteration determined to be present in the determining E), whether the respective genomic alteration is germline or somatic, and wherein when the respective genomic alteration is determined to be germline, suppressing use of the respective genomic alteration in providing the molecular residual disease (MRD) status or the cancer recurrence status for the human subject.
**11**. The computer system of claim 1, the method further comprising: storing the structured data report in a subject data store.
**12**. The computer system of claim 1, the method further comprising: retrieving health data associated with the human subject, and augmenting the structured data report with the health data.
**13**. The computer system of claim 12, wherein the health data comprises a cancer type associated with the human subject, and the augmenting the structured data report with the health data comprises including the cancer type in the data report.
**14**. The computer system of claim 12, wherein the health data comprises details of a specimen collected from the human subject, and the augmenting the data report with the health data comprises including details of the specimen in the data report.
**15**. The computer system of claim 12, wherein the health data comprises a diagnosis, a recurrence, a medication, a surgery, a response to treatment incurred by the human subject, an adverse effect to treatment incurred by the human subject, or an organoid modeling result, and the augmenting the data report comprises including the diagnosis, date of recurrence, the treatment incurred by the human subject, or an outcome of treatment in the data report.
**16**. The computer system of claim 12, wherein the health data is retrieved from an electronic medical record or an electronic health record associated with the human subject.
**17**. The computer system of claim 12, wherein the health data provides an indication of a status of a colorectal, ovarian, lung, or breast cancer for the human subject.
**18**. The computer system of claim 1, wherein the human subject is afflicted with colorectal, ovarian, lung, or breast cancer.
**19**. The computer system of claim 1, wherein the D) mapping is performed by a first microservice and the E) determining is performed by a second microservice, the method further comprising: publishing, upon completion of the D) mapping, the locations in the reference human genome to which the second genomic sequencing data have been mapped to a lake database, adding a first data alert or first event to a data alerts list, and initiating the second microservice to perform the E) determining upon the adding of the first data alert or first event to a data alerts list, wherein the E) determining obtains the locations in the reference human genome to which the second genomic sequencing data have been mapped from the lake database.
**20**. A non-transitory computer readable storage medium storing one or more programs, the one or more programs comprising instructions that, when executed by an electronic device with one or more processors and a memory, cause the electronic device to perform a method for evaluating a cancer treatment efficacy for a human subject that is undergoing, or has completed, a cancer treatment for a cancer, the method comprising: A) obtaining, from an electronic data store, first genomic sequencing data from cell free DNA drawn from a blood sample of the human subject, wherein the first genomic sequencing data comprises a plurality of epigenetic patterns of cell free DNA within the blood sample; B) obtaining, from an electronic data store, second genomic sequencing data from cell free DNA drawn from the blood sample, wherein the second genomic sequencing data comprises sequence data of cell free DNA within the blood sample; C) mapping the first genomic sequencing data to corresponding locations in a reference human genome; D) mapping the second genomic sequencing data to corresponding locations in the reference human genome; E) determining an absence or presence of at least a first genomic alteration, from among a plurality of predefined genomic alterations, in the blood sample based on at least (i) the second

genomic sequencing data, and (ii) the locations in the reference human genome to which the second genomic sequencing data has been mapped; and F) securely communicating, through a network connection, a structured data report that provides an assessment of the cancer treatment efficacy for the human subject responsive to at least the plurality of epigenetic patterns of cell free DNA within the blood sample and the absence or presence of at least the first genomic alteration in the blood sample.